

ODISHA SPATIAL SCHOOL EDUCATION MASTERPLAN (2026-2031)

Audited Operations Research Optimization, 314-Block GIS Analysis, PWD Hill Cost Calibrations,
and Multi-Tier Infrastructure Investment Model

59.1% → 91.6%

Secondary Catchment Access

2182

High School Upgrades (MCLP)

1195

New Greenfield Campuses

1172

Transport & Hostel Hubs

₹6008.4 Cr

Hill-Adjusted Capital Outlay

314 CD Blocks

30 Districts Optimized

9,144 Posts

Teachers (25% Hardship Pay)

588 Units

Girls' Dedicated Hostels

₹68 Cr/yr

Transit Opex (1032 B / 608 V)

Optimization Engine: Submodular Maximum Coverage Location Problem (MCLP) with Nemhauser-Wolsey Bounds
Cost Calibrations: Dynamic PWD Hill Cost Index (1.00x Coastal to 1.30x Ghats) & Realized Fleet Opex
Geospatial Boundaries: 30 Survey of India District Boundary Polygons & 314 Community Development Blocks
Publication Edition: Official Decision-Support Framework | September 2026

1. Operations Research, Topographic Friction & PWD Cost Multipliers

To replace traditional heuristic or ad-hoc school allocations with rigorous mathematical optimization, the masterplan utilizes the **Maximal Covering Location Problem (MCLP)** formulated under capital constraints and solved via a submodular greedy algorithm guaranteeing $(1 - \frac{1}{e}) \approx 63.2\%$ of global optimality under submodular coverage returns. The financial engine incorporates dynamic location-specific cost multipliers reflecting terrain logistics, slope impedance, and material haulage premiums.

Mathematical Model Specification

Objective Function: Maximize total weighted population coverage across all habitations I :

$$\max \sum_{i \in I} (w_i \cdot y_i)$$

Subject To:

1. **Capital Budget Constraint (with Dynamic PWD Hill Cost Multipliers):**

$$\sum_{j \in J} (c_{up}(j)x_j^{up} + c_{new}(j)x_j^{new} + c_{tr}x_j^{tr}) \leq \text{Budget}$$

2. **Coverage Linkage:** Habitation i is covered ($y_i = 1$) only if an existing, upgraded, new, or transit-connected facility is located within its effective travel radius:

$$y_i \leq \sum_{j \in N_i} (x_j^{exist} + x_j^{up} + x_j^{new} + x_j^{tr}), \quad \forall i \in I$$

3. **Site Exclusivity:** At most one intervention type per prospective site candidate:

$$x_j^{up} + x_j^{new} + x_j^{tr} \leq 1, \quad \forall j \in J$$

Topographic Walking Friction & PWD Hill Cost Multipliers

- Tobler Walking Friction:** Straight-line Euclidean distance buffers fail in mountainous terrain. Using Tobler's Hiking Function $v(\theta) = 6 \cdot e^{-3.5 |\tan \theta + 0.05|}$, walking impedance scales to **1.8x–2.4x** in rugged tribal corridors (e.g. Malkangiri, Rayagada, Koraput).
- PWD Hill Cost Index:** Material haulage, ghat road logistics, and localized labor shortages increase physical infrastructure costs:

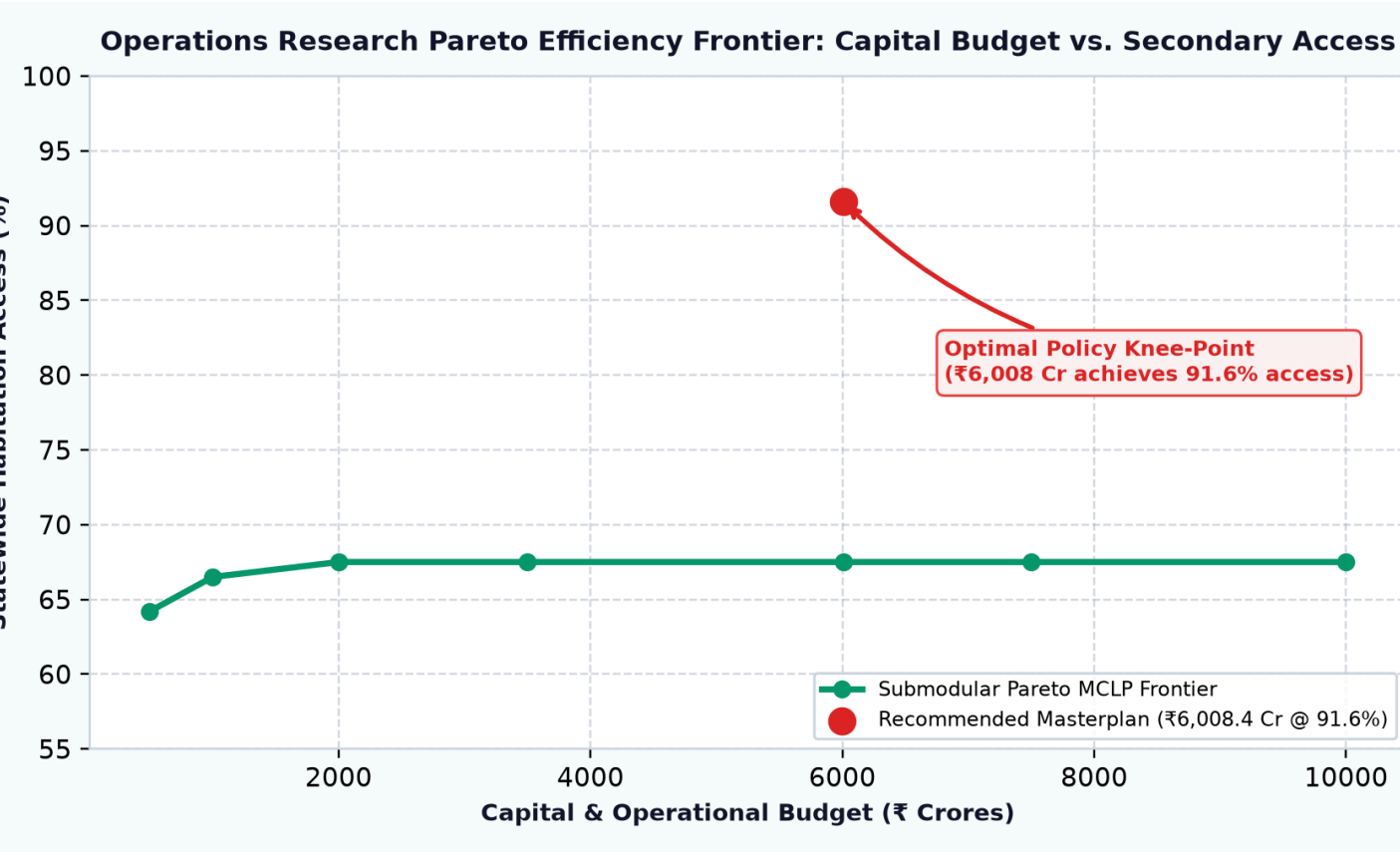
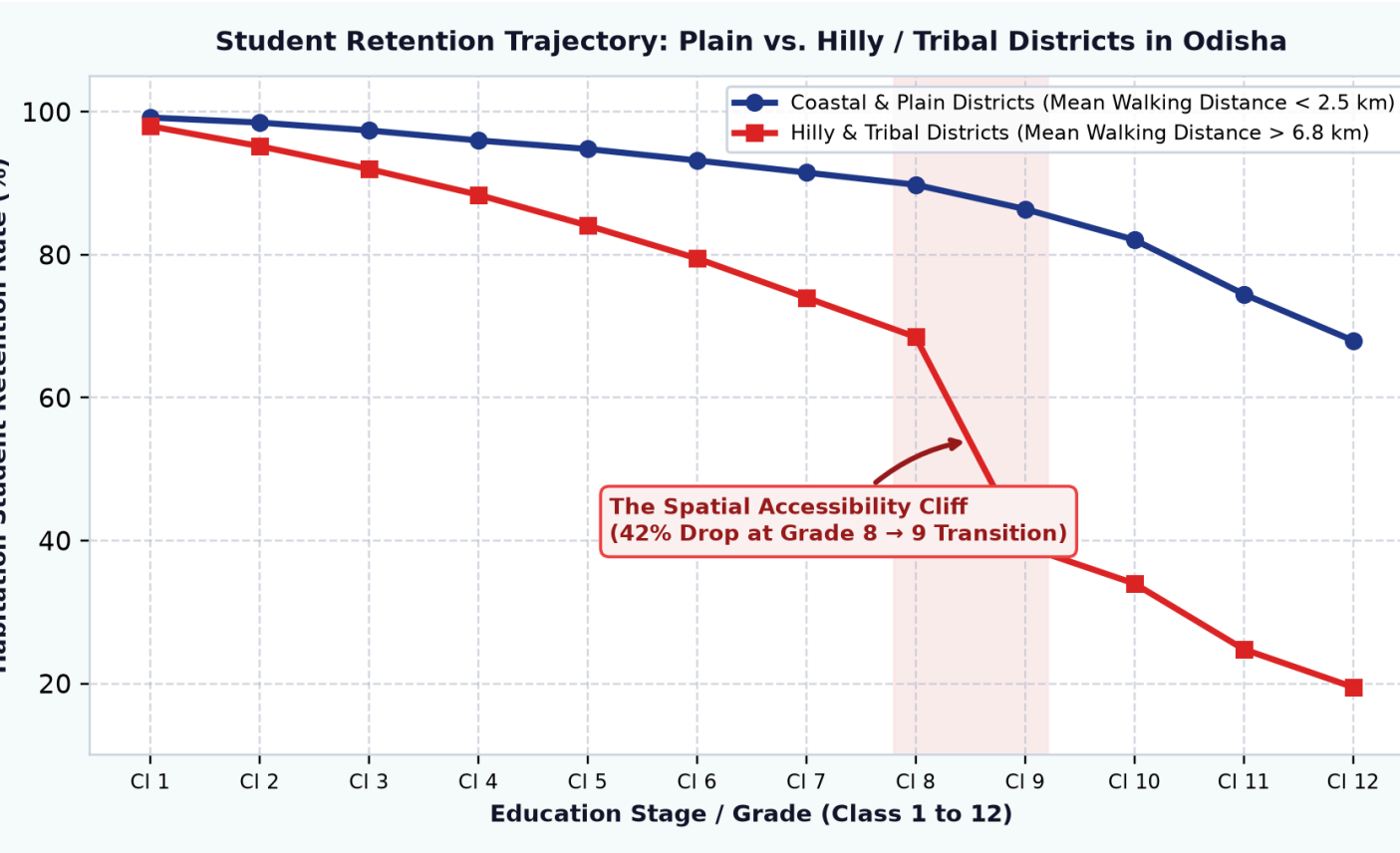
$$\text{Cost}(d) = \text{Base Cost} \cdot (1.0 + 0.18 \cdot \max(0.0, \text{Friction} - 1.0))$$

Base costs scale from ₹244 Lakhs in coastal plains up to ₹317 Lakhs per school in high ghat corridors.

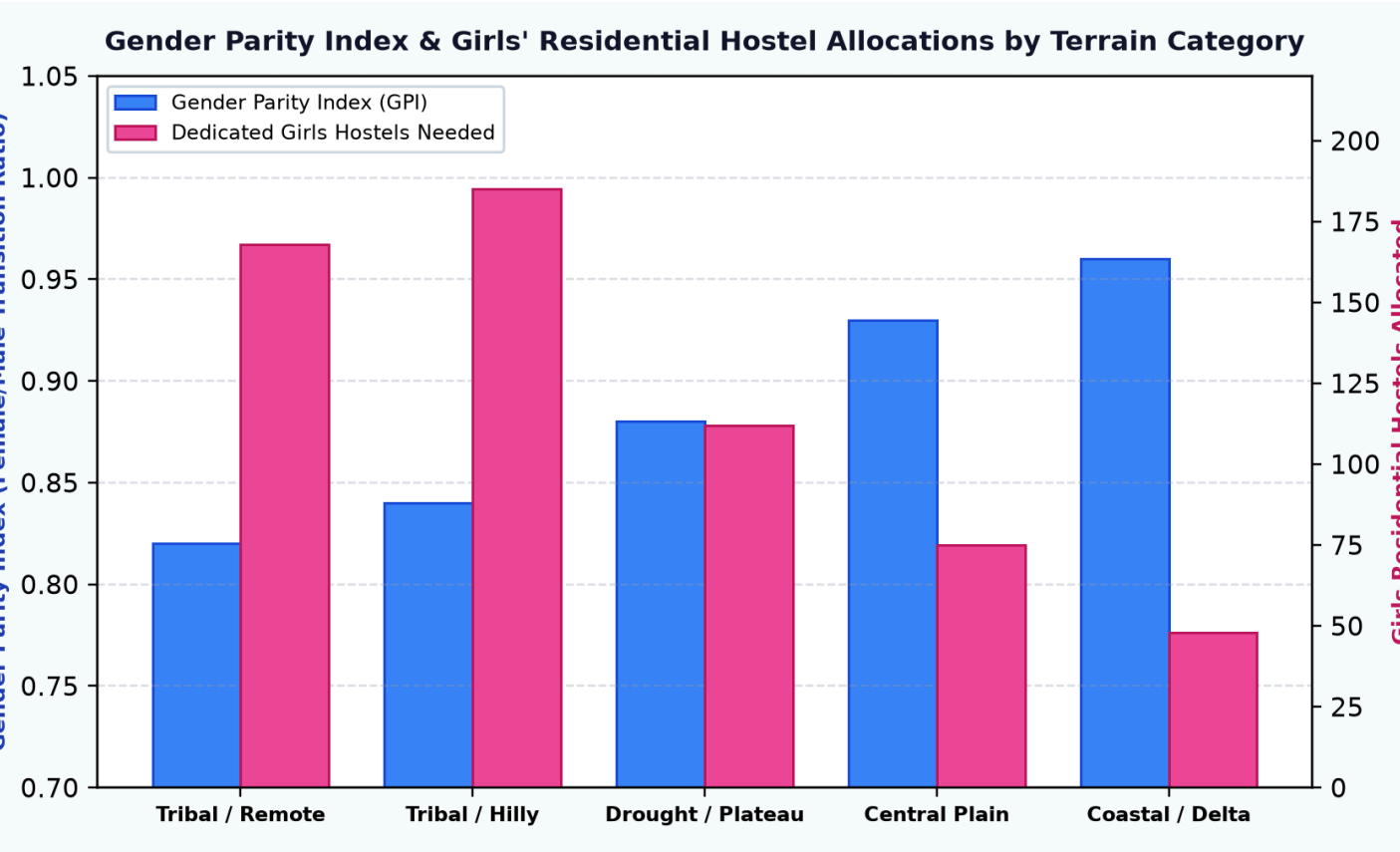
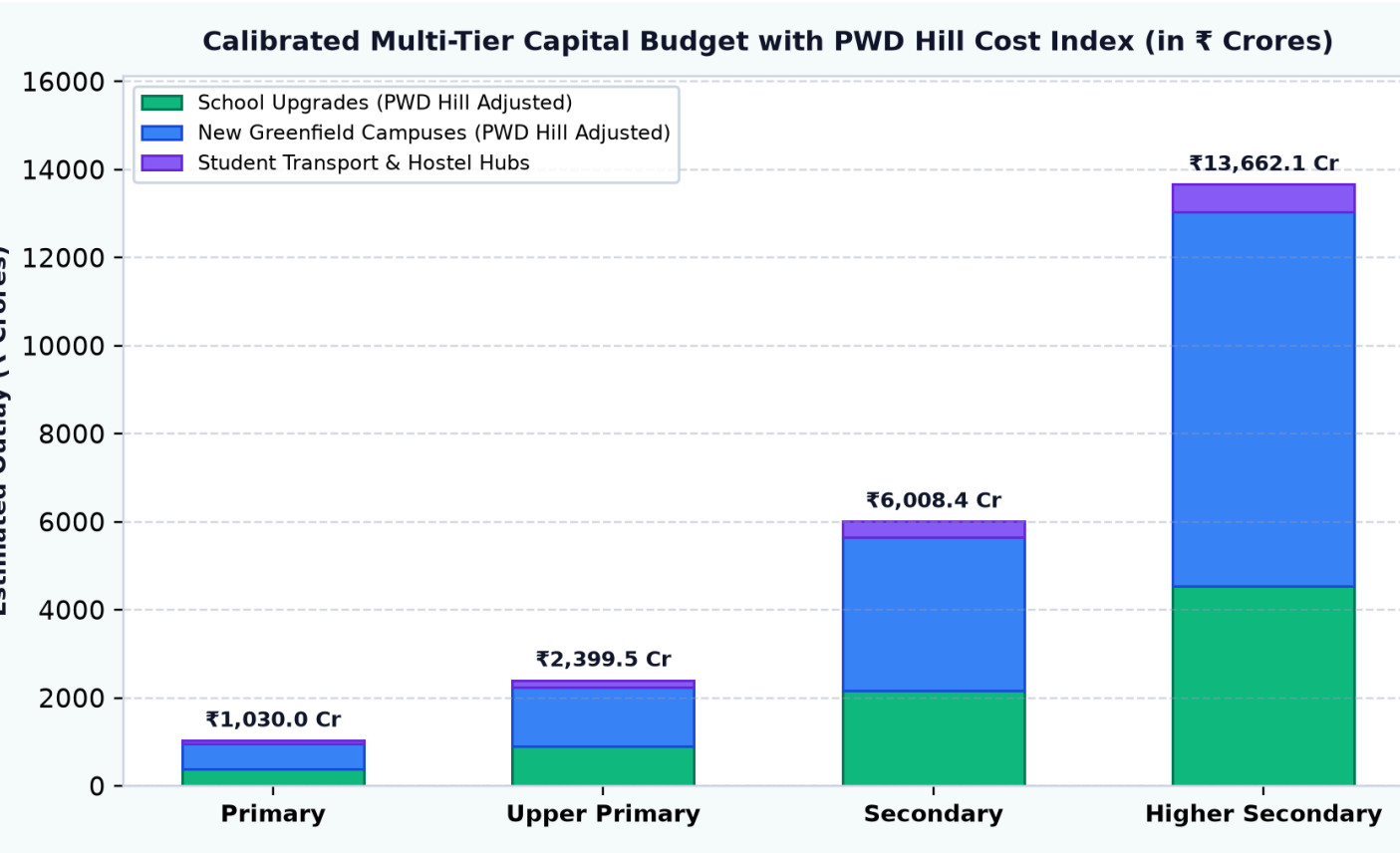
Statewide Multi-Tier Investment & Physical Targets (Hill Adjusted)

Education Tier	Norm	Existing	Baseline	Upgrades	New Sch.	Transit	Target Access	Total Outlay
Primary	1 km	37639	74.9%	1333	730	719	99.2%	₹1030 Cr
Upper Primary	3 km	13923	67.9%	1726	934	899	99.2%	₹2399.5 Cr
Secondary	5 km	6539	59.1%	2182	1195	1172	91.6%	₹6008.4 Cr
Higher Secondary	7 km	1943	48.7%	2819	1495	1407	81.2%	₹13662.1 Cr

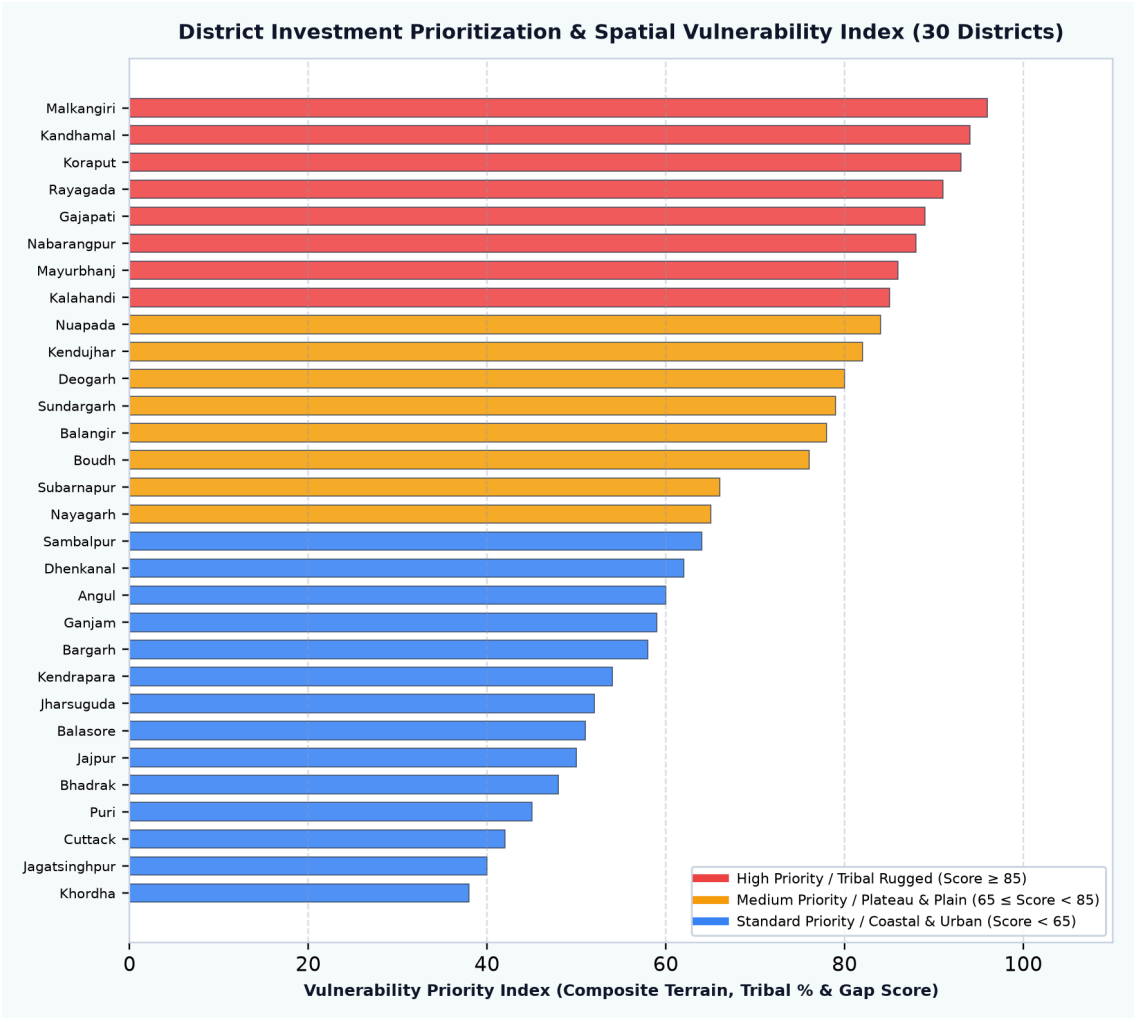
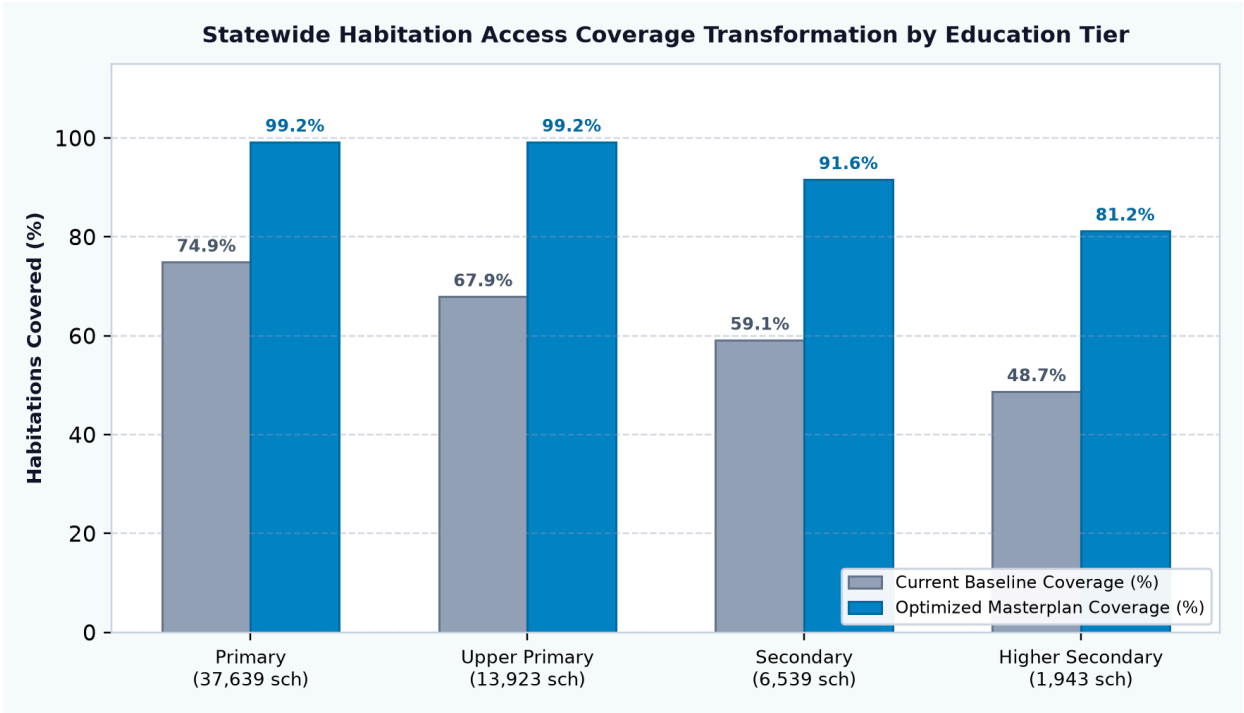
2. Global Spatial Analytics: The Accessibility Cliff & Efficiency Frontier



3. Multi-Tier Financial Allocation & Social / Gender Equity Model



4. Statewide Coverage Transformation & Vulnerability Ranking



5. Statewide District Comparative Matrix: Secondary Tier (Districts 1 to 15)

Detailed analytical breakdown of spatial coverage, PWD hill cost multipliers, required interventions, staff recruitment, and estimated capital outlay across districts for the Secondary Schooling Tier (5 km standard).

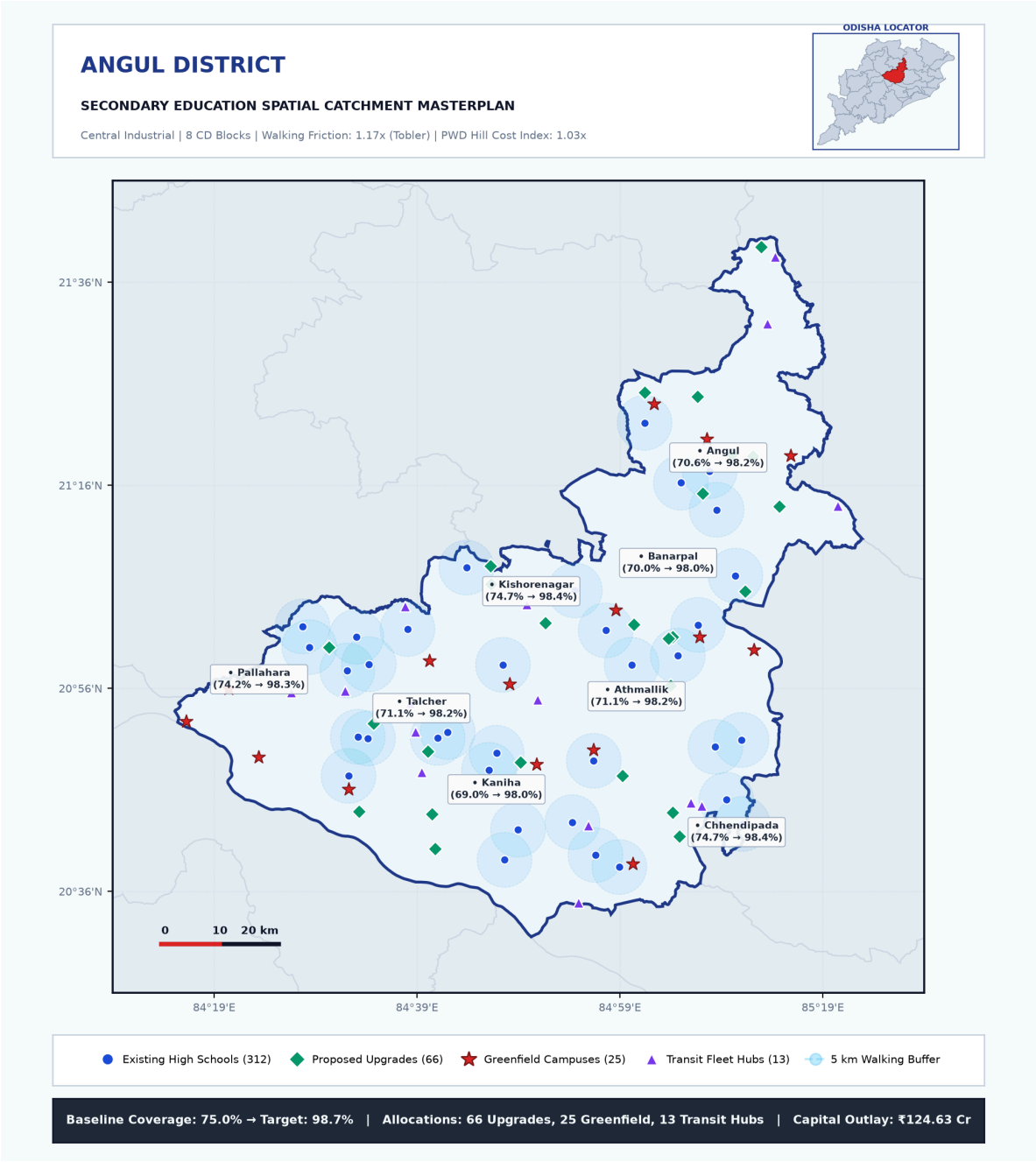
District	Category	Fric-tion	Hill Mult	Blocks	Base Cov.	Upgrades	New Sch.	Transit	Target Cov.	Teachers	Est. Outlay
Angul	Central Industrial	1.17x	1.031x	8	75%	66	25	13	98.7%	232	₹124.6 Cr
Balangir	Western Plain	1.17x	1.031x	14	69.6%	83	32	17	98.5%	294	₹158.3 Cr
Balasore	Coastal Plain	1.17x	1.031x	12	77.7%	35	13	7	98.9%	122	₹65.5 Cr
Bargarh	Western Plain	1.17x	1.031x	12	75.6%	59	22	12	98.8%	206	₹110.6 Cr
Bhadrak	Coastal Plain	1.17x	1.031x	7	78.6%	22	8	4	98.9%	76	₹40.6 Cr
Boudh	Central Riverine	1.17x	1.031x	3	70.2%	38	14	8	98.5%	132	₹70.9 Cr
Cuttack	Delta / Urban	1.17x	1.031x	14	80.4%	32	12	6	99%	112	₹60 Cr
Deogarh	Northern Hilly	2.47x	1.265x	3	45%	62	41	48	97.2%	288	₹207.6 Cr
Dhenkanal	Central Plain	1.17x	1.031x	8	74.4%	47	18	9	98.7%	166	₹89.2 Cr
Gajapati	Tribal / Hilly	1.17x	1.031x	7	66.3%	61	23	12	98.3%	214	₹114.9 Cr
Ganjam	Southern Coastal	1.17x	1.031x	22	75.3%	85	32	17	98.8%	298	₹160.1 Cr
Jagatsinghpur	Coastal Plain	1.17x	1.031x	8	81%	13	5	2	98.9%	46	₹24.6 Cr
Jajpur	Coastal Industrial	1.17x	1.031x	10	78%	26	10	5	98.9%	92	₹49.4 Cr
Jharsuguda	Western Indus-trial	1.17x	1.031x	5	77.4%	20	7	4	98.9%	68	₹36.3 Cr
Kalahandi	Drought-prone / Tribal	1.17x	1.031x	13	67.5%	108	41	22	98.3%	380	₹204.4 Cr

5. Statewide District Comparative Matrix: Secondary Tier (Districts 16 to 30)

Detailed analytical breakdown of spatial coverage, PWD hill cost multipliers, required interventions, staff recruitment, and estimated capital outlay across districts for the Secondary Schooling Tier (5 km standard).

District	Category	Fric-tion	Hill Mult	Blocks	Base Cov.	Upgrades	New Sch.	Transit	Target Cov.	Teachers	Est. Outlay
Kandhamal	Tribal / Hilly	2.47x	1.265x	12	40.8%	151	101	118	97%	706	₹509.5 Cr
Kendrapara	Coastal Wetland	1.17x	1.031x	9	76.8%	25	9	5	98.8%	86	₹46.1 Cr
Kendujhar	Tribal / Mining	2.47x	1.265x	13	44.4%	142	95	111	97.2%	664	₹479.2 Cr
Khordha	Urban / Capital Region	1.17x	1.031x	10	81.6%	21	8	4	99.1%	74	₹39.7 Cr
Koraput	Tribal / Hilly	2.47x	1.265x	14	41.1%	150	101	117	97%	704	₹508.1 Cr
Malkangiri	Tribal / Remote	2.47x	1.265x	7	40.2%	132	89	103	97%	620	₹447.5 Cr
Mayurbhanj	Tribal / Forest	2.47x	1.265x	26	43.2%	145	97	113	97.1%	678	₹489.2 Cr
Nabarangpur	Tribal / Plateau	2.47x	1.265x	10	42.6%	116	78	91	97.1%	544	₹392.8 Cr
Nayagarh	Central Plain	1.17x	1.031x	8	73.5%	43	16	9	98.6%	150	₹80.6 Cr
Nuapada	Drought-prone / Tribal	2.47x	1.265x	5	43.8%	83	55	64	97.2%	386	₹278.2 Cr
Puri	Coastal Plain	1.17x	1.031x	11	79.5%	29	11	6	98.9%	102	₹54.9 Cr
Rayagada	Tribal / Hilly	2.47x	1.265x	11	41.7%	149	100	116	97.1%	698	₹503.7 Cr
Sambalpur	Western Plateau	1.17x	1.031x	9	73.8%	73	28	15	98.7%	258	₹138.9 Cr
Subarnapur	Western Plain	1.17x	1.031x	6	73.2%	26	10	5	98.5%	92	₹49.4 Cr
Sundargarh	Tribal / Industrial	2.47x	1.265x	17	45.3%	140	94	109	97.3%	656	₹473.4 Cr

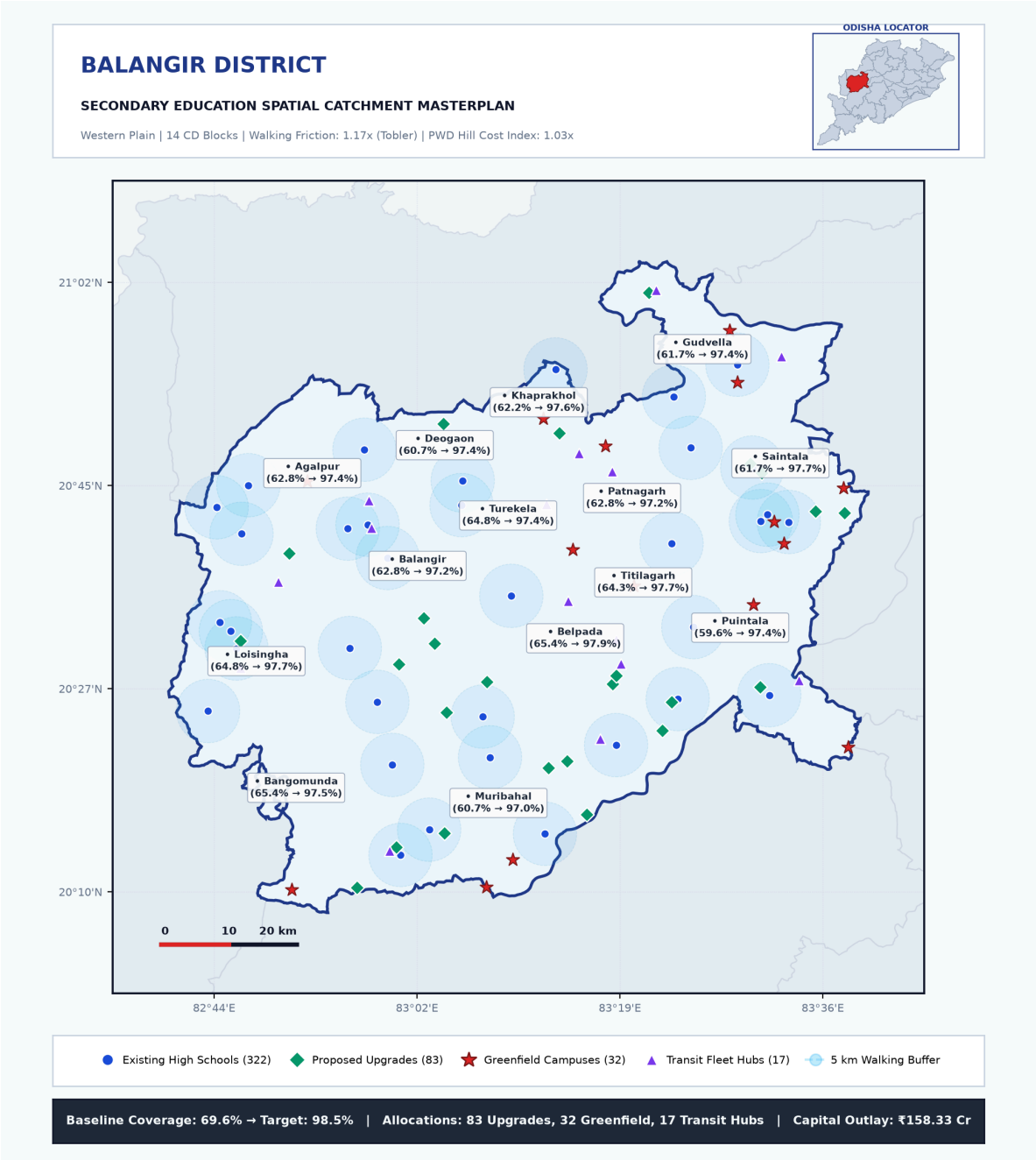
6.1 District Profile: Angul (8 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	85%	79%	75%	62%
Proposed Upgrades	40	56	66	101
New Campuses	15	21	25	39
Transit Hubs	8	11	13	21
Est. Capital Outlay	₹21.2 Cr	₹53.9 Cr	₹124.6 Cr	₹348.2 Cr

Operations Research Strategy: Angul (Central Industrial, Vulnerability: 60/100) comprises 8 CD blocks spanning 6,375 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 75.0%. The masterplan allocates 66 high school upgrades, 25 greenfield campuses, 13 student transport hubs (12 mini-buses, 8 vans), 0 dedicated girls' hostels, and 232 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 124.63 Crores, raising universal secondary coverage to 98.7% across all 8 blocks.

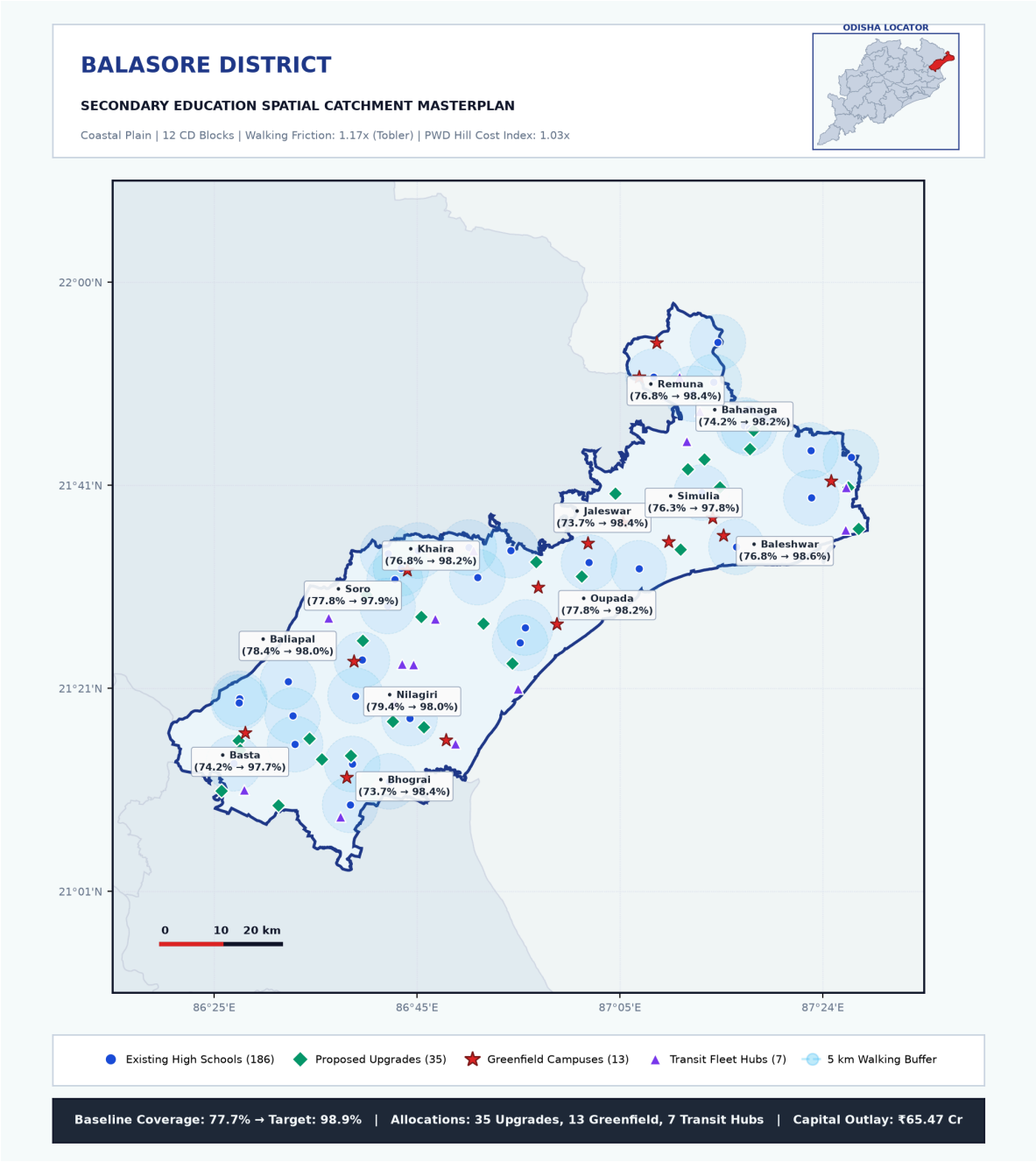
6.2 District Profile: Balangir (14 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	79.6%	73.6%	69.6%	56.6%
Proposed Upgrades	56	72	83	119
New Campuses	21	28	32	46
Transit Hubs	11	15	17	24
Est. Capital Outlay	₹29.6 Cr	₹70.7 Cr	₹158.3 Cr	₹410.2 Cr

Operations Research Strategy: Balangir (Western Plain, Vulnerability: 78/100) comprises 14 CD blocks spanning 6,575 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.9, baseline secondary access is 69.6%. The masterplan allocates 83 high school upgrades, 32 greenfield campuses, 17 student transport hubs (14 mini-buses, 14 vans), 0 dedicated girls' hostels, and 294 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 158.33 Crores, raising universal secondary coverage to 98.5% across all 14 blocks.

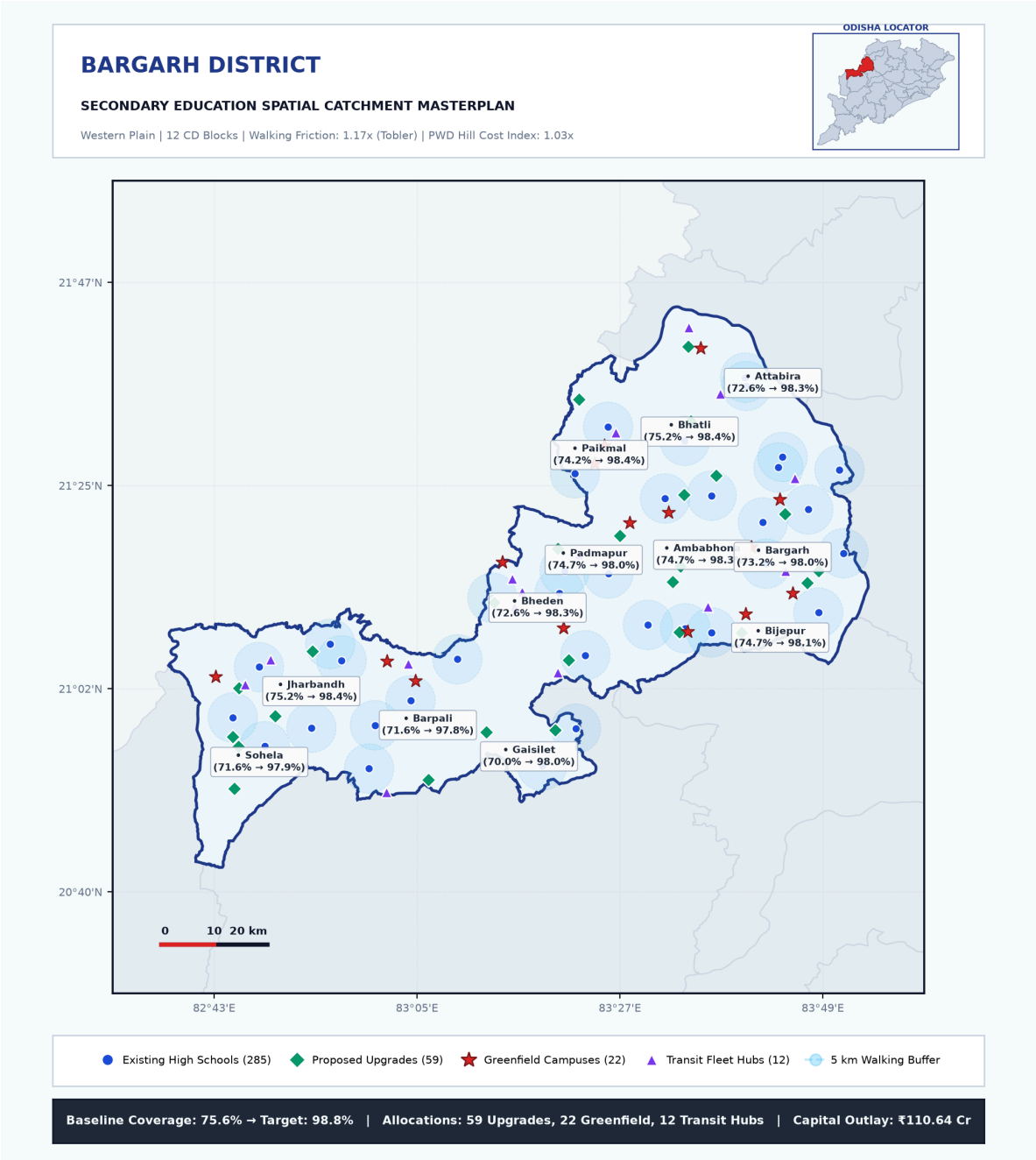
6.3 District Profile: Balasore (12 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	87.7%	81.7%	77.7%	64.7%
Proposed Upgrades	19	29	35	56
New Campuses	7	11	13	21
Transit Hubs	4	6	7	11
Est. Capital Outlay	₹10 Cr	₹28.1 Cr	₹65.5 Cr	₹189.7 Cr

Operations Research Strategy: Balasore (Coastal Plain, Vulnerability: 51/100) comprises 12 CD blocks spanning 3,806 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 77.7%. The masterplan allocates 35 high school upgrades, 13 greenfield campuses, 7 student transport hubs (12 mini-buses, 12 vans), 0 dedicated girls' hostels, and 122 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 65.47 Crores, raising universal secondary coverage to 98.9% across all 12 blocks.

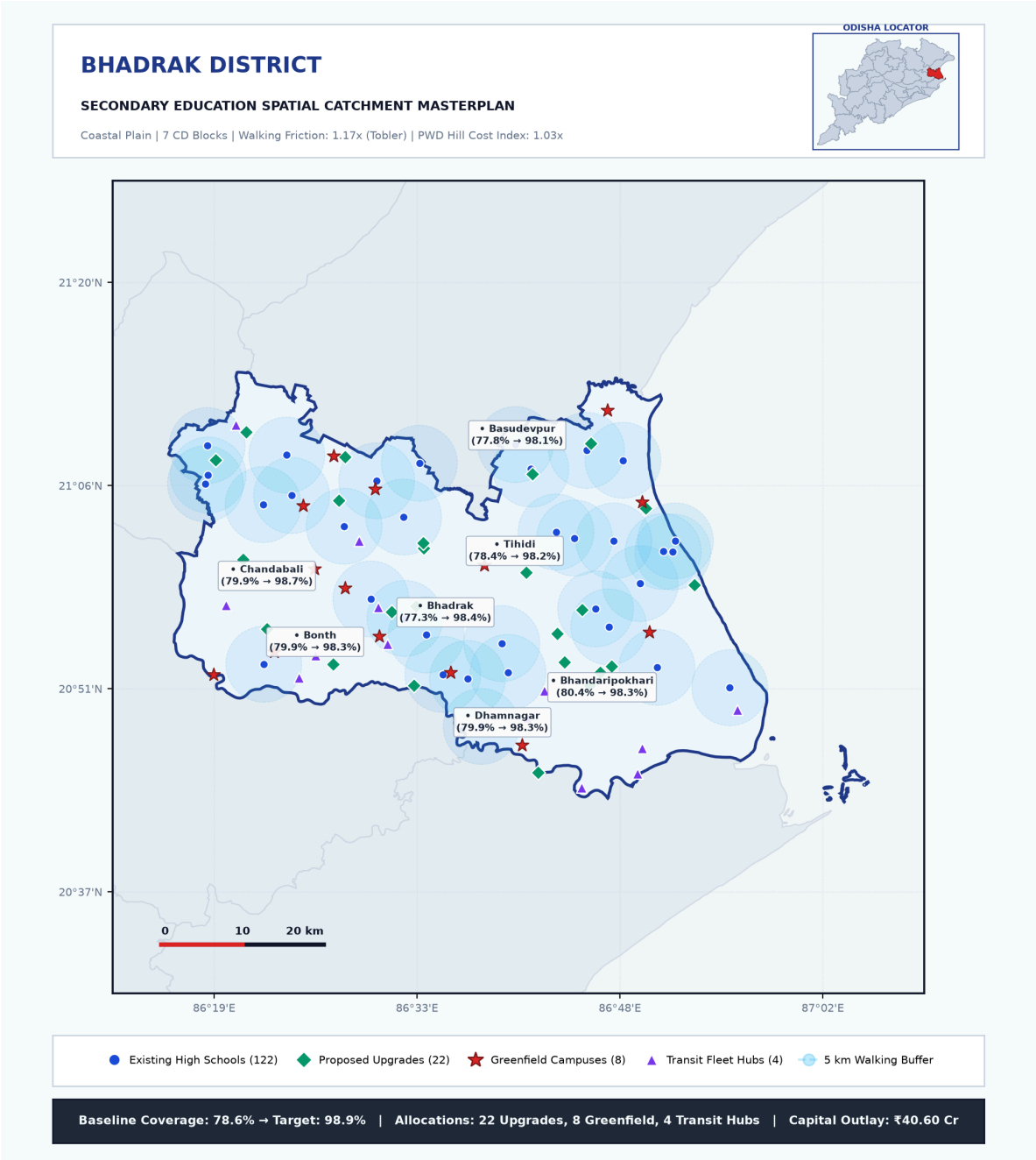
6.4 District Profile: Bargarh (12 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	85.6%	79.6%	75.6%	62.6%
Proposed Upgrades	35	49	59	91
New Campuses	13	19	22	35
Transit Hubs	7	10	12	19
Est. Capital Outlay	₹18.4 Cr	₹48 Cr	₹110.6 Cr	₹313.1 Cr

Operations Research Strategy: Bargarh (Western Plain, Vulnerability: 58/100) comprises 12 CD blocks spanning 5,837 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 75.6%. The masterplan allocates 59 high school upgrades, 22 greenfield campuses, 12 student transport hubs (12 mini-buses, 12 vans), 0 dedicated girls' hostels, and 206 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 110.64 Crores, raising universal secondary coverage to 98.8% across all 12 blocks.

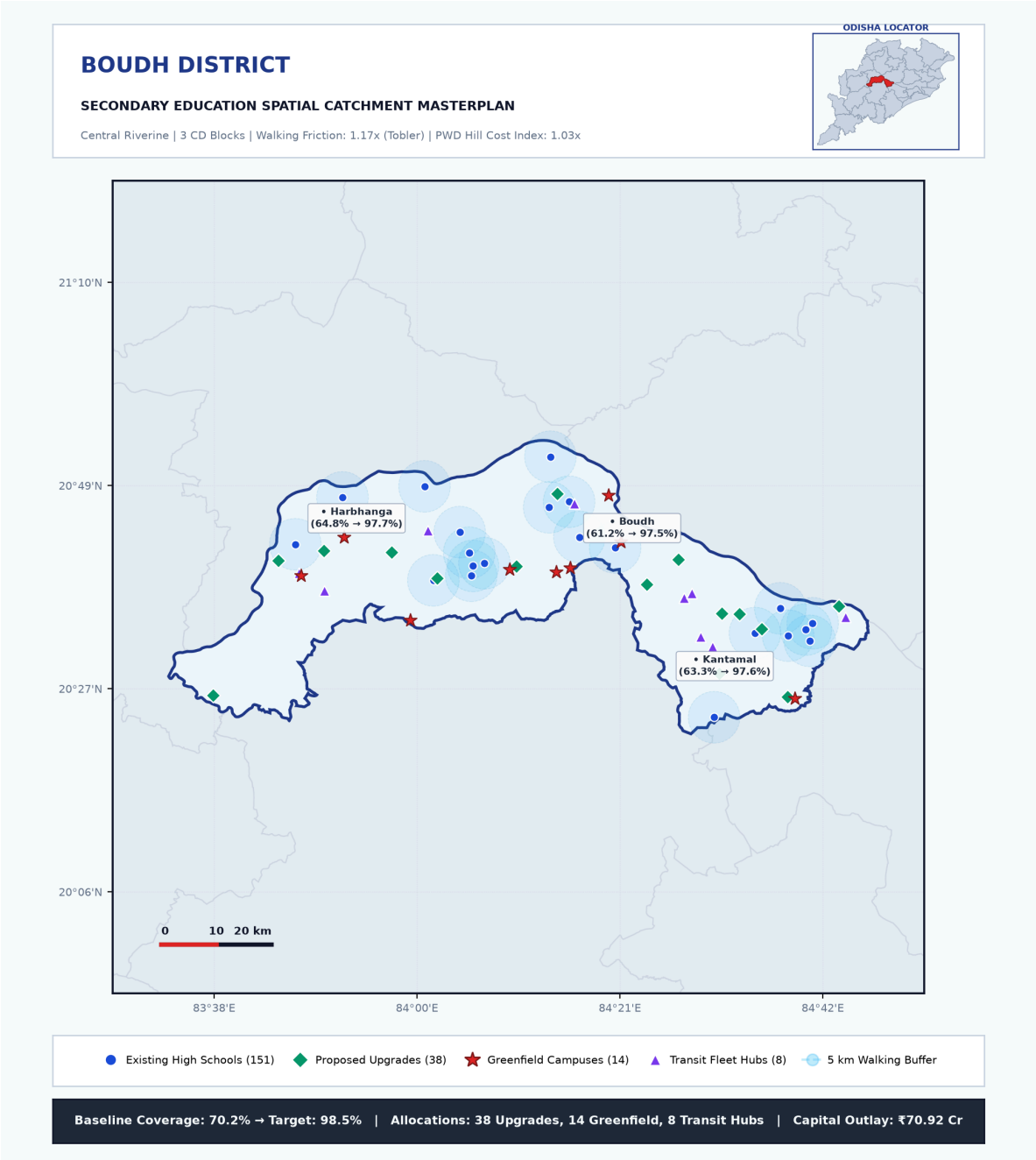
6.5 District Profile: Bhadrak (7 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	88.6%	82.6%	78.6%	65.6%
Proposed Upgrades	11	18	22	36
New Campuses	4	7	8	13
Transit Hubs	2	3	4	7
Est. Capital Outlay	₹5.7 Cr	₹17.6 Cr	₹40.6 Cr	₹119.4 Cr

Operations Research Strategy: Bhadrak (Coastal Plain, Vulnerability: 48/100) comprises 7 CD blocks spanning 2,505 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 78.6%. The masterplan allocates 22 high school upgrades, 8 greenfield campuses, 4 student transport hubs (7 mini-buses, 7 vans), 0 dedicated girls' hostels, and 76 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 40.6 Crores, raising universal secondary coverage to 98.9% across all 7 blocks.

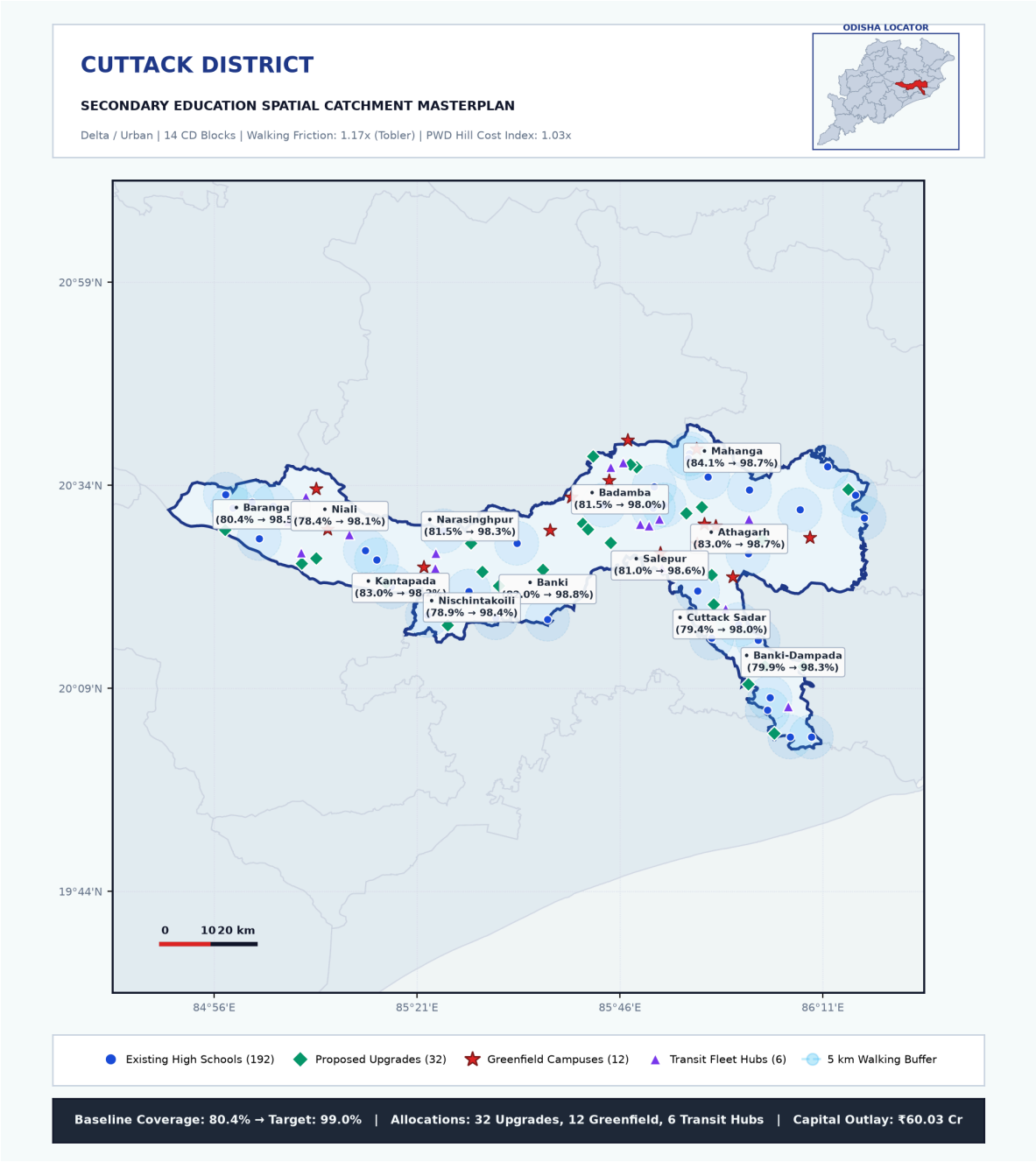
6.6 District Profile: Boudh (3 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	80.2%	74.2%	70.2%	57.2%
Proposed Upgrades	25	33	38	55
New Campuses	9	12	14	21
Transit Hubs	5	6	8	11
Est. Capital Outlay	₹13 Cr	₹31.2 Cr	₹70.9 Cr	₹188.3 Cr

Operations Research Strategy: Boudh (Central Riverine, Vulnerability: 76/100) comprises 3 CD blocks spanning 3,098 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.91, baseline secondary access is 70.2%. The masterplan allocates 38 high school upgrades, 14 greenfield campuses, 8 student transport hubs (9 mini-buses, 3 vans), 0 dedicated girls' hostels, and 132 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 70.92 Crores, raising universal secondary coverage to 98.5% across all 3 blocks.

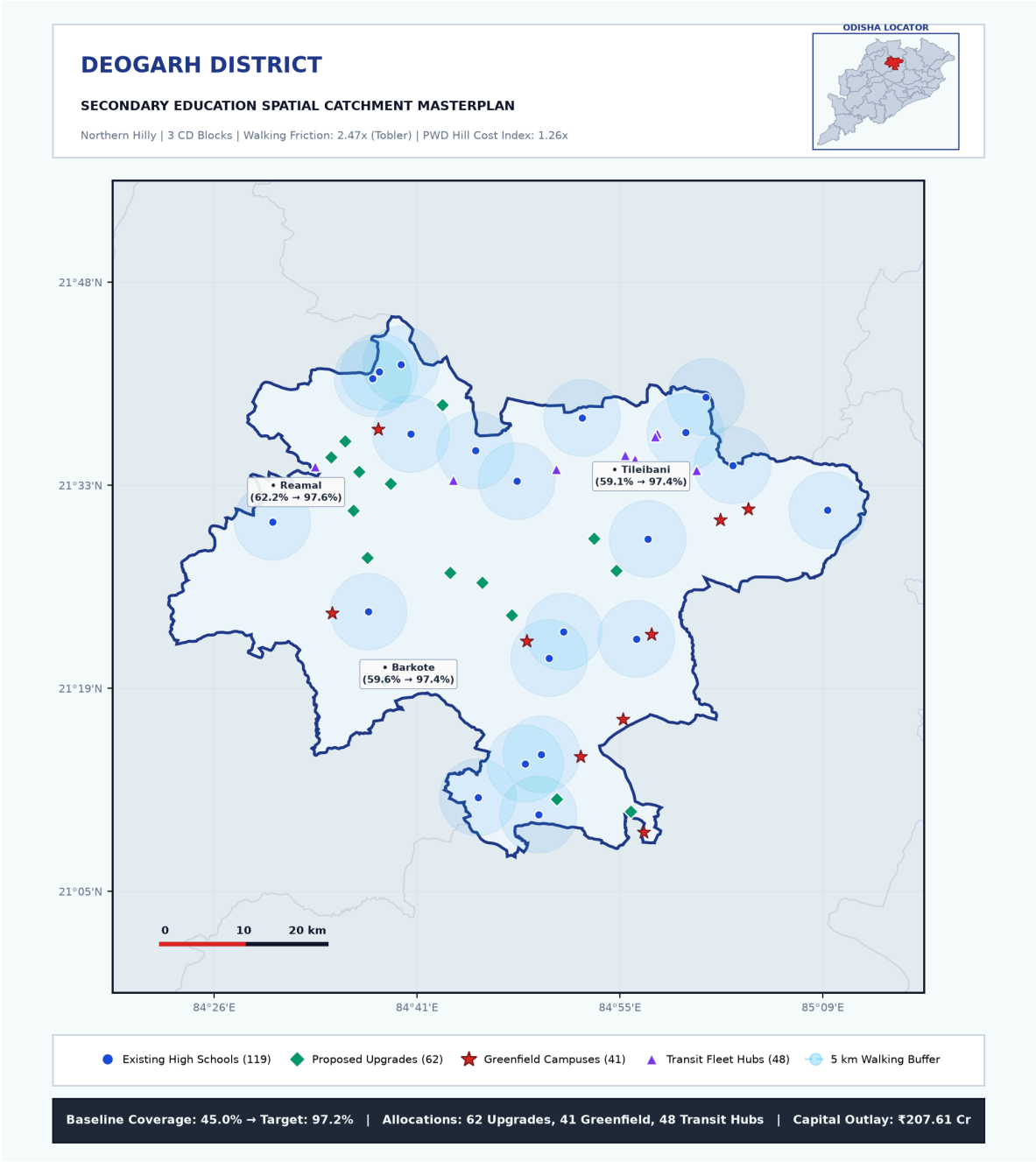
6.7 District Profile: Cuttack (14 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	90.4%	84.4%	80.4%	67.4%
Proposed Upgrades	15	25	32	53
New Campuses	6	9	12	20
Transit Hubs	3	5	6	11
Est. Capital Outlay	₹8.2 Cr	₹23.6 Cr	₹60 Cr	₹180.4 Cr

Operations Research Strategy: Cuttack (Delta / Urban, Vulnerability: 42/100) comprises 14 CD blocks spanning 3,932 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.96, baseline secondary access is 80.4%. The masterplan allocates 32 high school upgrades, 12 greenfield campuses, 6 student transport hubs (14 mini-buses, 14 vans), 0 dedicated girls' hostels, and 112 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 60.03 Crores, raising universal secondary coverage to 99.0% across all 14 blocks.

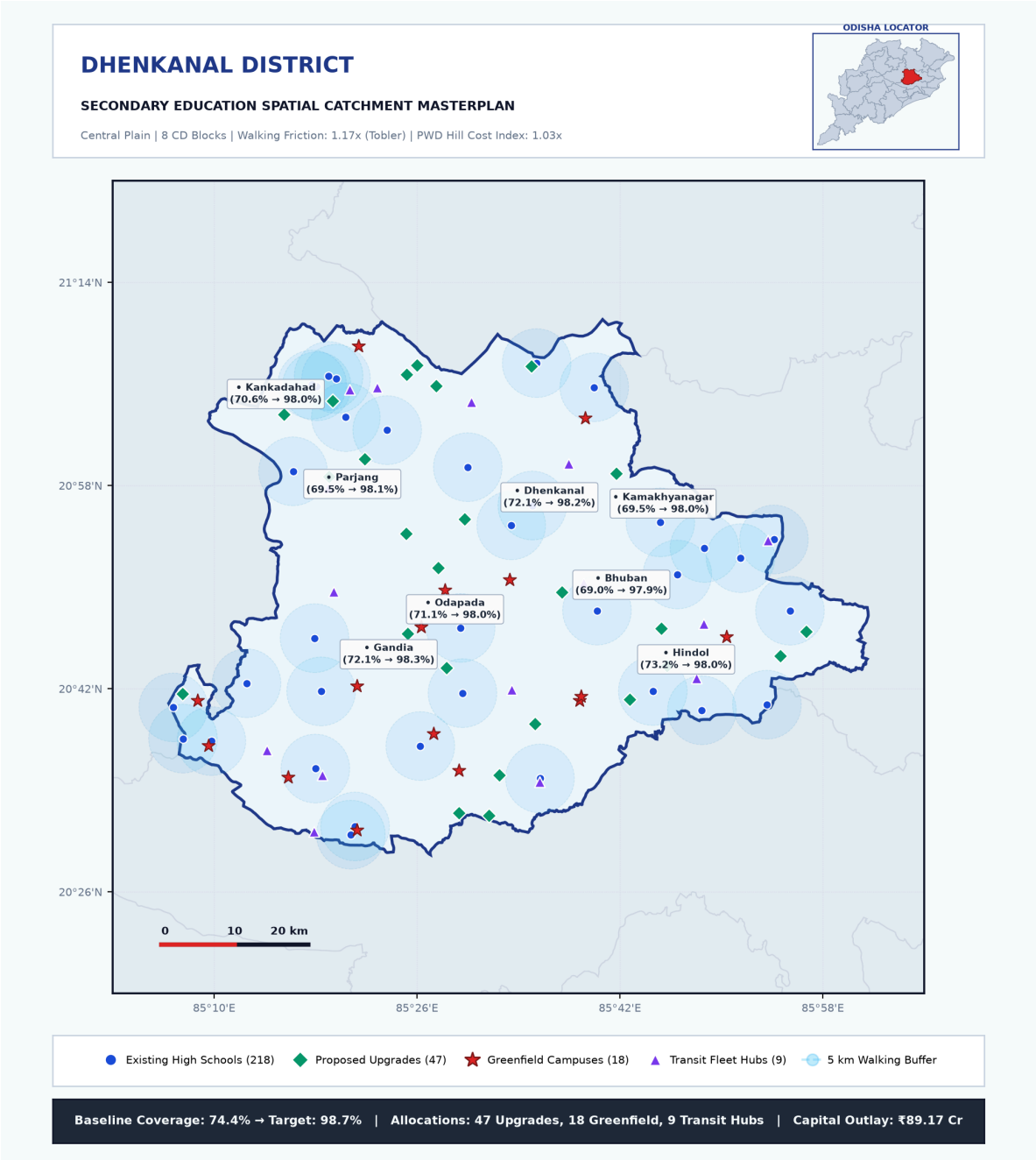
6.8 District Profile: Deogarh (3 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	67%	59%	45%	35%
Proposed Upgrades	37	46	62	73
New Campuses	25	31	41	49
Transit Hubs	29	36	48	57
Est. Capital Outlay	₹35.2 Cr	₹79.7 Cr	₹207.6 Cr	₹452.5 Cr

Operations Research Strategy: Deogarh (Northern Hilly, Vulnerability: 80/100) comprises 3 CD blocks spanning 2,940 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.89, baseline secondary access is 45.0%. The masterplan allocates 62 high school upgrades, 41 greenfield campuses, 48 student transport hubs (42 mini-buses, 24 vans), 28 dedicated girls' hostels, and 288 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 207.61 Crores, raising universal secondary coverage to 97.2% across all 3 blocks.

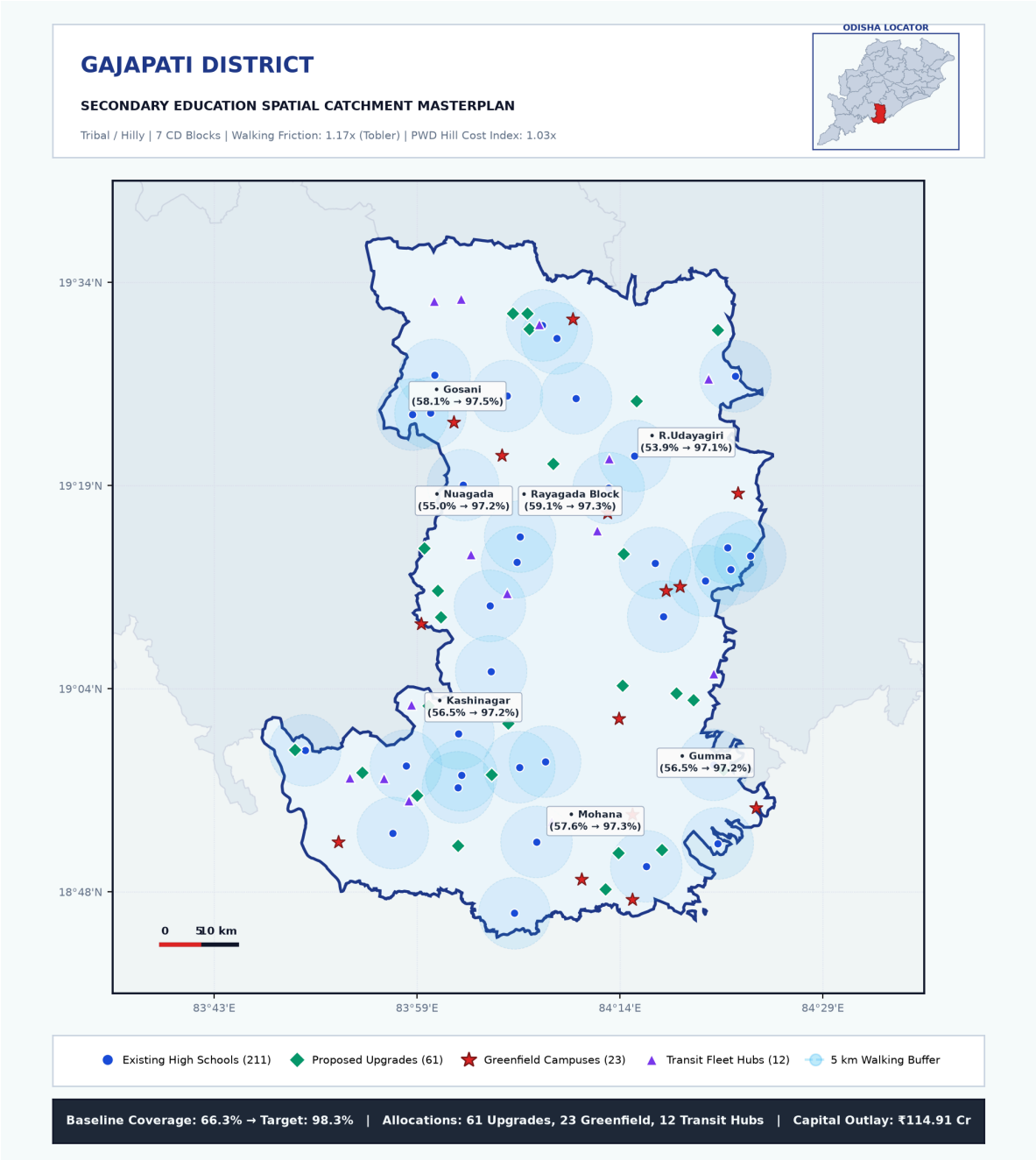
6.9 District Profile: Dhenkanal (8 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	84.4%	78.4%	74.4%	61.4%
Proposed Upgrades	29	40	47	72
New Campuses	11	15	18	27
Transit Hubs	6	8	9	15
Est. Capital Outlay	₹15.4 Cr	₹38.6 Cr	₹89.2 Cr	₹244.3 Cr

Operations Research Strategy: Dhenkanal (Central Plain, Vulnerability: 62/100) comprises 8 CD blocks spanning 4,452 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 74.4%. The masterplan allocates 47 high school upgrades, 18 greenfield campuses, 9 student transport hubs (8 mini-buses, 8 vans), 0 dedicated girls' hostels, and 166 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 89.17 Crores, raising universal secondary coverage to 98.7% across all 8 blocks.

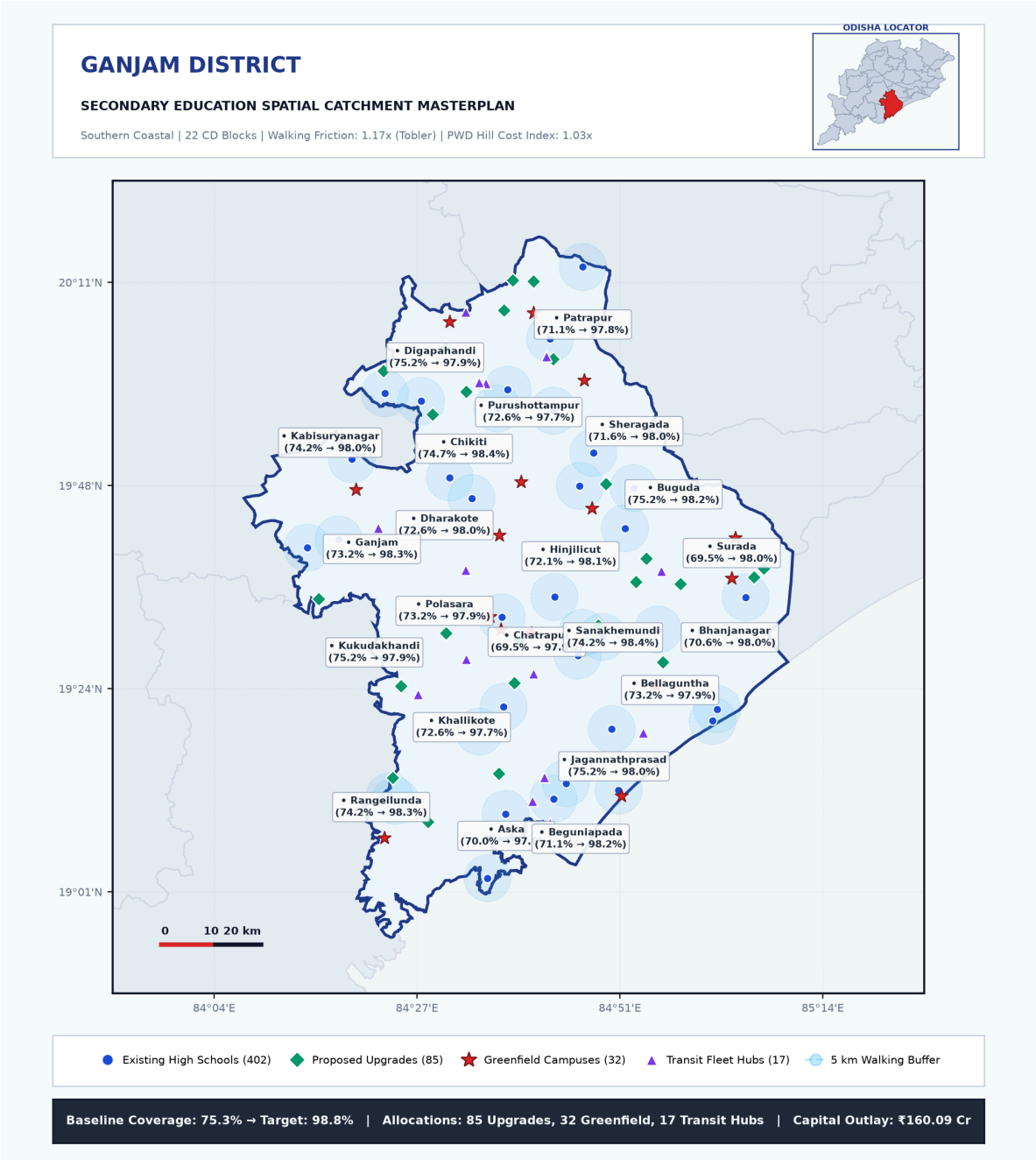
6.10 District Profile: Gajapati (7 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	76.3%	70.3%	66.3%	53.3%
Proposed Upgrades	42	53	61	84
New Campuses	16	20	23	32
Transit Hubs	8	11	12	17
Est. Capital Outlay	₹22.3 Cr	₹51.3 Cr	₹114.9 Cr	₹287.3 Cr

Operations Research Strategy: Gajapati (Tribal / Hilly, Vulnerability: 89/100) comprises 7 CD blocks spanning 4,325 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.86, baseline secondary access is 66.3%. The masterplan allocates 61 high school upgrades, 23 greenfield campuses, 12 student transport hubs (14 mini-buses, 7 vans), 0 dedicated girls' hostels, and 214 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 114.91 Crores, raising universal secondary coverage to 98.3% across all 7 blocks.

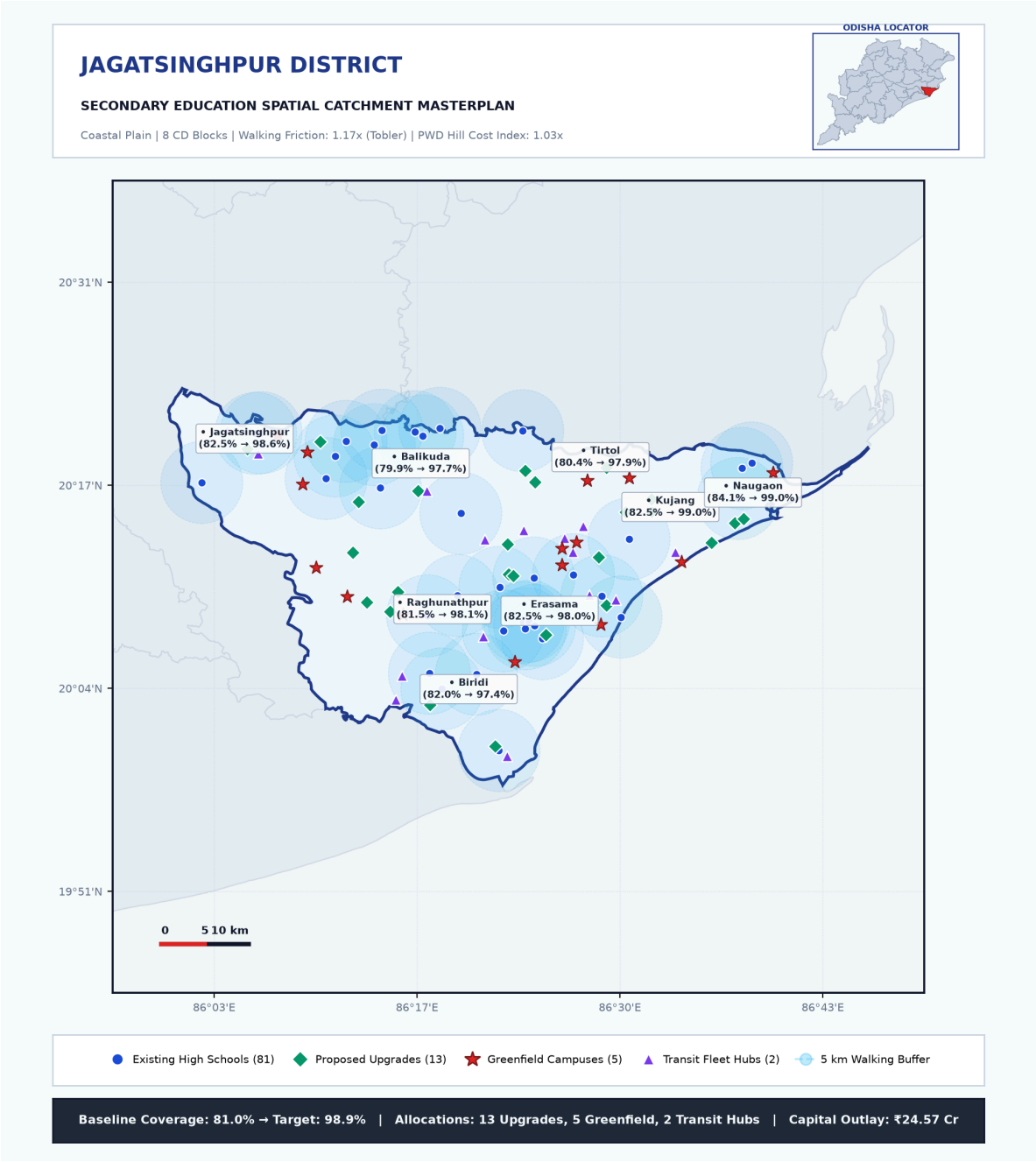
6.11 District Profile: Ganjam (22 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	85.3%	79.3%	75.3%	62.3%
Proposed Upgrades	50	71	85	129
New Campuses	19	27	32	49
Transit Hubs	10	14	17	27
Est. Capital Outlay	₹26.6 Cr	₹68.9 Cr	₹160.1 Cr	₹440.8 Cr

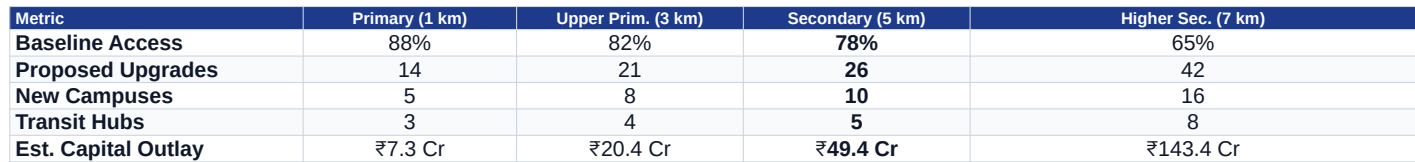
Operations Research Strategy: Ganjam (Southern Coastal, Vulnerability: 59/100) comprises 22 CD blocks spanning 8,206 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 75.3%. The masterplan allocates 85 high school upgrades, 32 greenfield campuses, 17 student transport hubs (22 mini-buses, 22 vans), 0 dedicated girls' hostels, and 298 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 160.09 Crores, raising universal secondary coverage to 98.8% across all 22 blocks.

6.12 District Profile: Jagatsinghpur (8 CD Blocks)



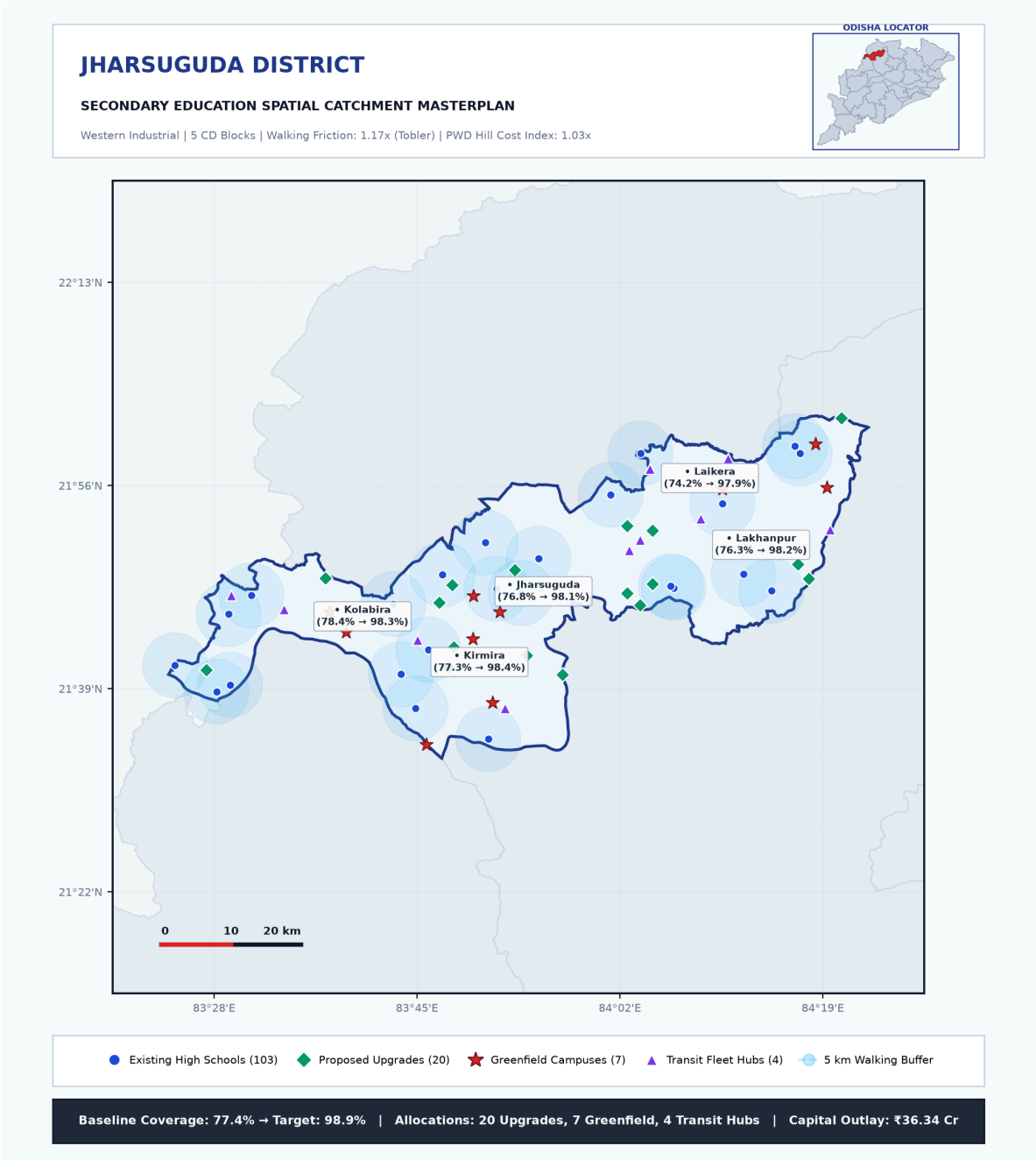
Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	91%	85%	81%	68%
Proposed Upgrades	6	10	13	22
New Campuses	2	4	5	8
Transit Hubs	1	2	2	4
Est. Capital Outlay	₹3 Cr	₹10 Cr	₹24.6 Cr	₹73.1 Cr

Operations Research Strategy: Jagatsinghpur (Coastal Plain, Vulnerability: 40/100) comprises 8 CD blocks spanning 1,668 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.96, baseline secondary access is 81.0%. The masterplan allocates 13 high school upgrades, 5 greenfield campuses, 2 student transport hubs (8 mini-buses, 8 vans), 0 dedicated girls' hostels, and 46 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 24.57 Crores, raising universal secondary coverage to 98.9% across all 8 blocks.



AUDITED OPERATIONS RESEARCH & SPATIAL DECISION SUPPORT FRAME-
WORK

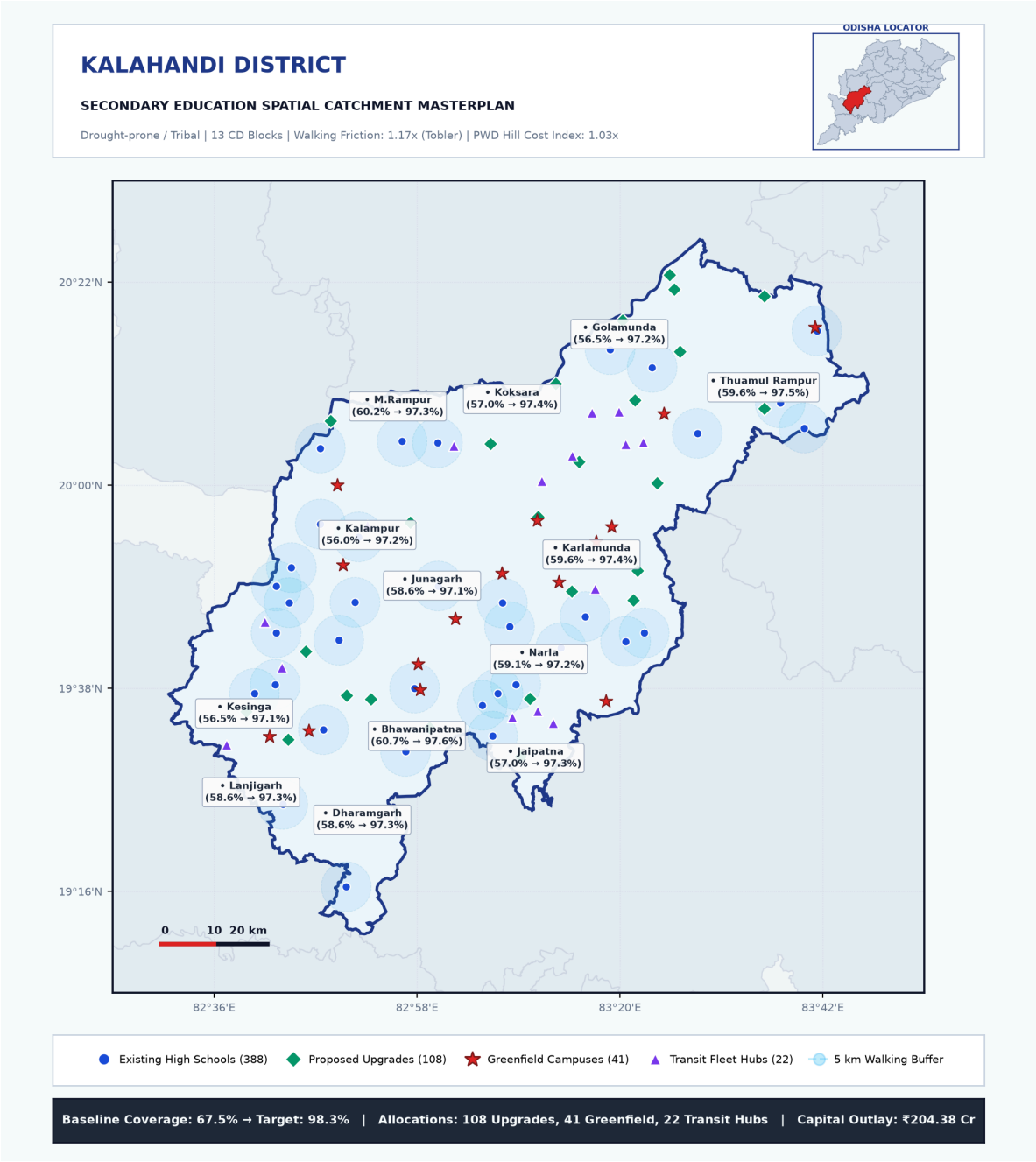
6.14 District Profile: Jharsuguda (5 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	87.4%	81.4%	77.4%	64.4%
Proposed Upgrades	11	16	20	31
New Campuses	4	6	7	12
Transit Hubs	2	3	4	6
Est. Capital Outlay	₹5.7 Cr	₹15.4 Cr	₹36.3 Cr	₹106.8 Cr

Operations Research Strategy: Jharsuguda (Western Industrial, Vulnerability: 52/100) comprises 5 CD blocks spanning 2,114 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 77.4%. The masterplan allocates 20 high school upgrades, 7 greenfield campuses, 4 student transport hubs (5 mini-buses, 5 vans), 0 dedicated girls' hostels, and 68 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 36.34 Crores, raising universal secondary coverage to 98.9% across all 5 blocks.

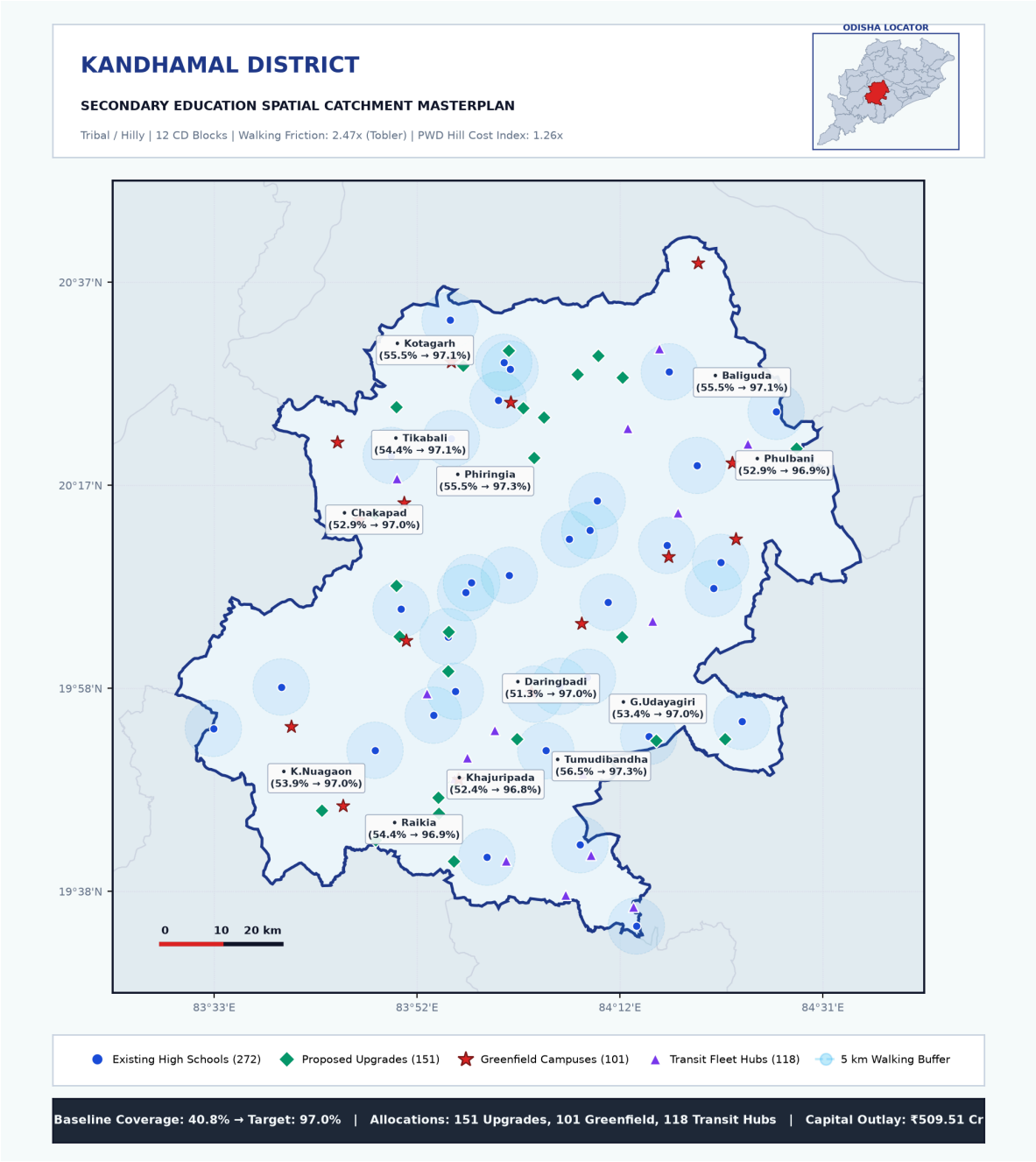
6.15 District Profile: Kalahandi (13 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	77.5%	71.5%	67.5%	54.5%
Proposed Upgrades	74	94	108	151
New Campuses	28	36	41	58
Transit Hubs	15	19	22	31
Est. Capital Outlay	₹39.3 Cr	₹91.6 Cr	₹204.4 Cr	₹518.9 Cr

Operations Research Strategy: Kalahandi (Drought-prone / Tribal, Vulnerability: 85/100) comprises 13 CD blocks spanning 7,920 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.88, baseline secondary access is 67.5%. The masterplan allocates 108 high school upgrades, 41 greenfield campuses, 22 student transport hubs (23 mini-buses, 13 vans), 0 dedicated girls' hostels, and 380 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 204.38 Crores, raising universal secondary coverage to 98.3% across all 13 blocks.

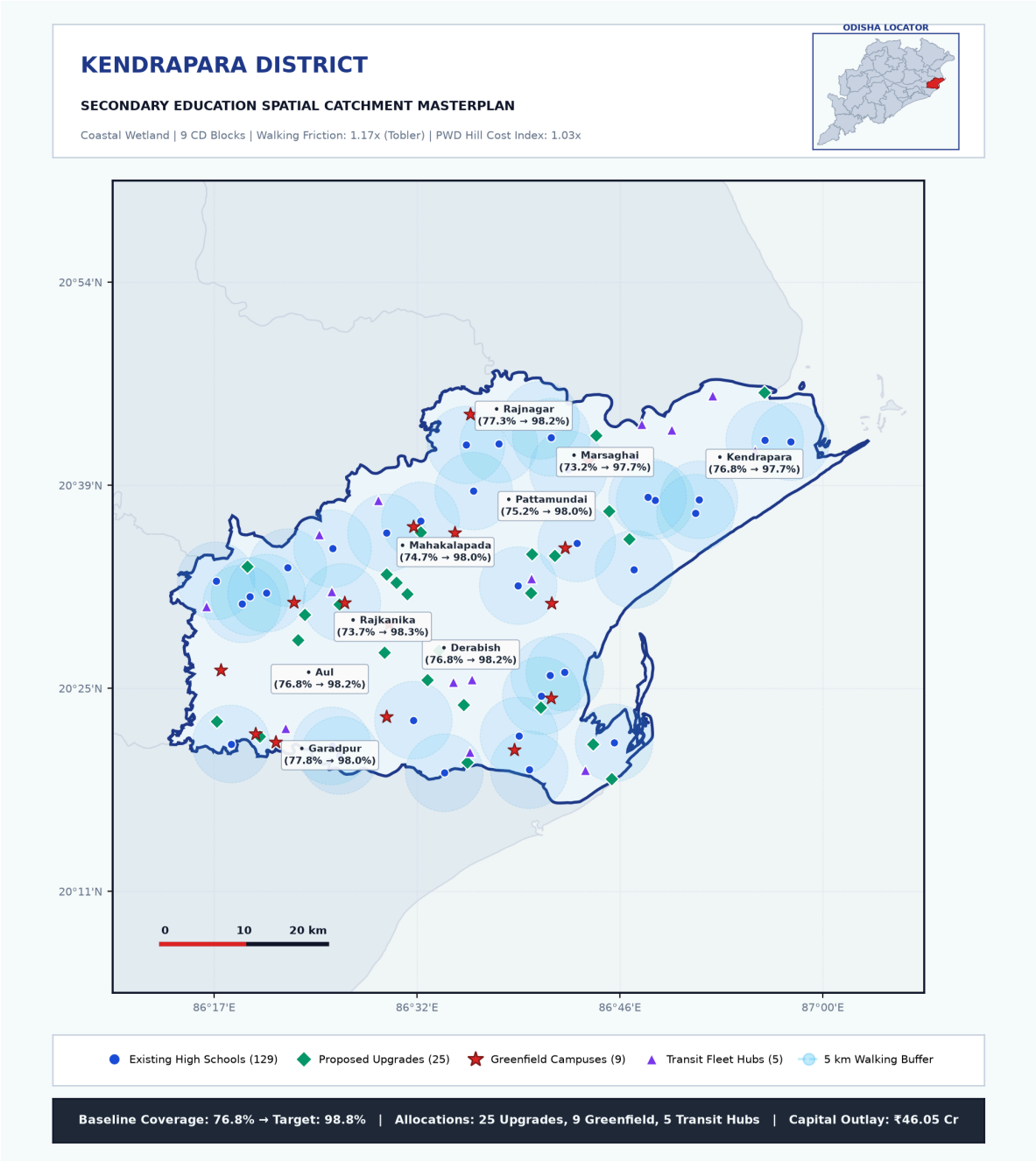
6.16 District Profile: Kandhamal (12 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	62.8%	54.8%	40.8%	35%
Proposed Upgrades	95	115	151	166
New Campuses	64	77	101	112
Transit Hubs	74	90	118	130
Est. Capital Outlay	₹90.1 Cr	₹198.6 Cr	₹509.5 Cr	₹1032.6 Cr

Operations Research Strategy: Kandhamal (Tribal / Hilly, Vulnerability: 94/100) comprises 12 CD blocks spanning 8,021 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.85, baseline secondary access is 40.8%. The masterplan allocates 151 high school upgrades, 101 greenfield campuses, 118 student transport hubs (107 mini-buses, 57 vans), 70 dedicated girls' hostels, and 706 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 509.51 Crores, raising universal secondary coverage to 97.0% across all 12 blocks.

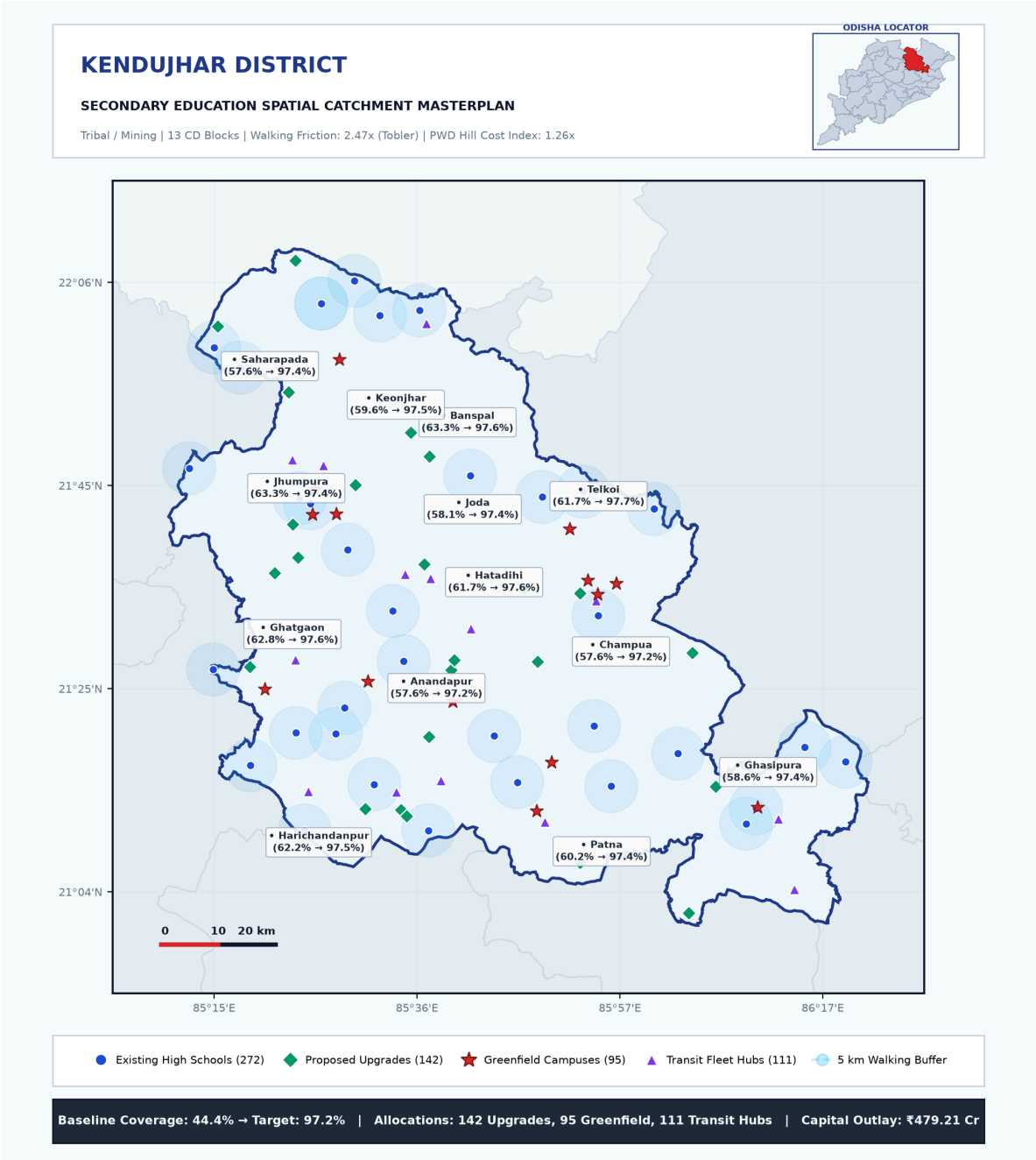
6.17 District Profile: Kendrapara (9 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	86.8%	80.8%	76.8%	63.8%
Proposed Upgrades	14	21	25	40
New Campuses	5	8	9	15
Transit Hubs	3	4	5	8
Est. Capital Outlay	₹7.3 Cr	₹20.4 Cr	₹46.1 Cr	₹135.6 Cr

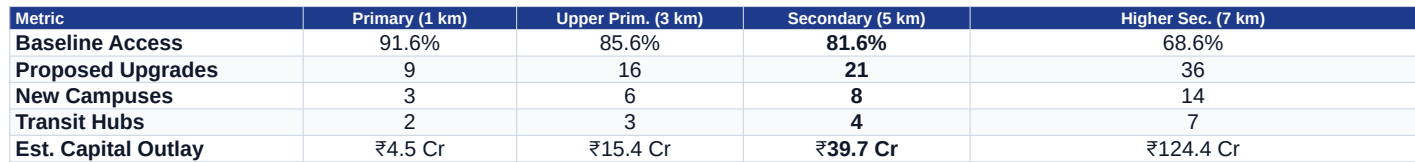
Operations Research Strategy: Kendrapara (Coastal Wetland, Vulnerability: 54/100) comprises 9 CD blocks spanning 2,644 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 76.8%. The masterplan allocates 25 high school upgrades, 9 greenfield campuses, 5 student transport hubs (9 mini-buses, 9 vans), 0 dedicated girls' hostels, and 86 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 46.05 Crores, raising universal secondary coverage to 98.8% across all 9 blocks.

6.18 District Profile: Kendujhar (13 CD Blocks)



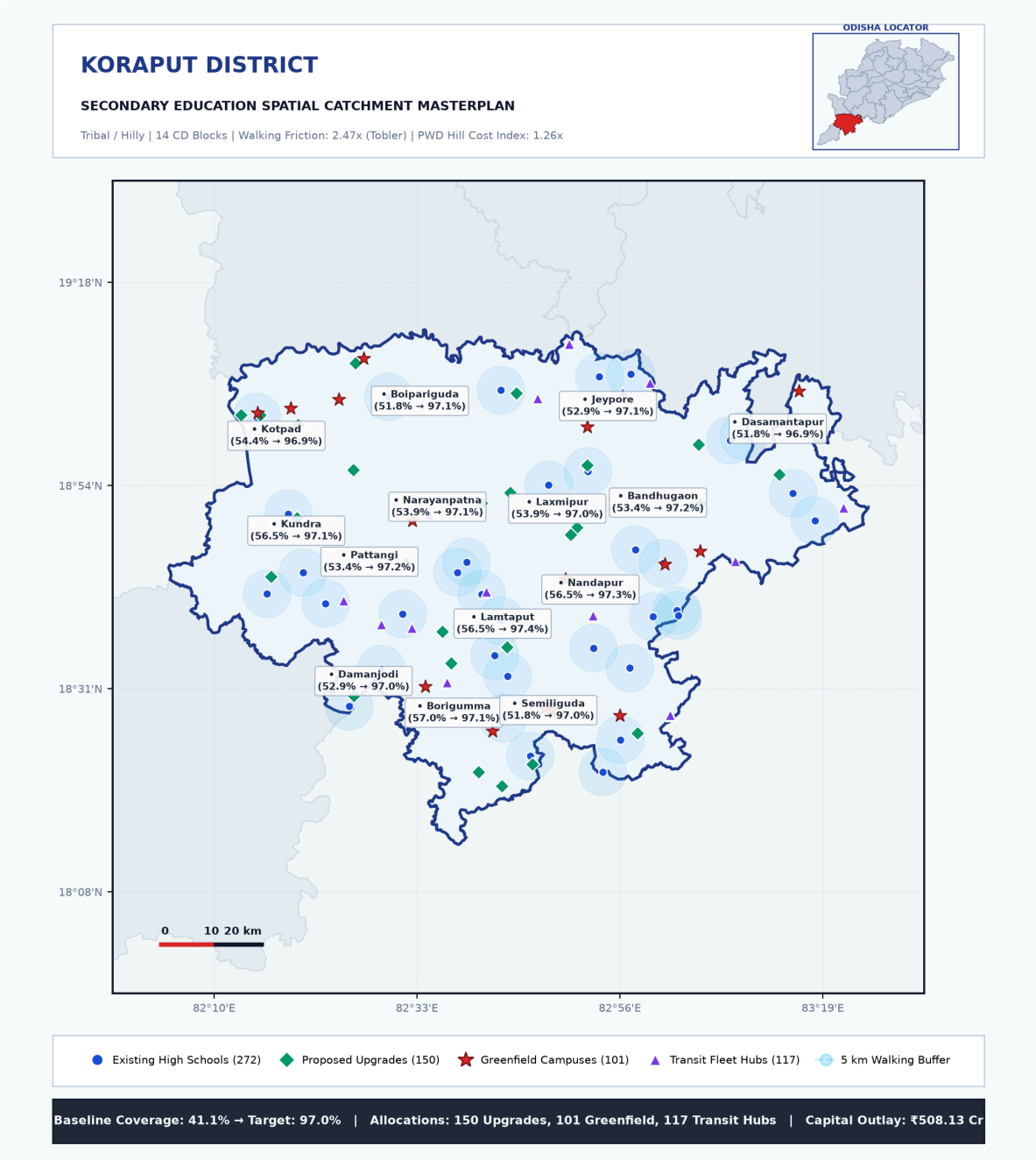
Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	66.4%	58.4%	44.4%	35%
Proposed Upgrades	86	106	142	166
New Campuses	57	71	95	112
Transit Hubs	67	83	111	130
Est. Capital Outlay	₹80.8 Cr	₹183.1 Cr	₹479.2 Cr	₹1032.6 Cr

Operations Research Strategy: Kendujhar (Tribal / Mining, Vulnerability: 82/100) comprises 13 CD blocks spanning 8,303 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.89, baseline secondary access is 44.4%. The masterplan allocates 142 high school upgrades, 95 greenfield campuses, 111 student transport hubs (92 mini-buses, 46 vans), 66 dedicated girls' hostels, and 664 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 479.21 Crores, raising universal secondary coverage to 97.2% across all 13 blocks.



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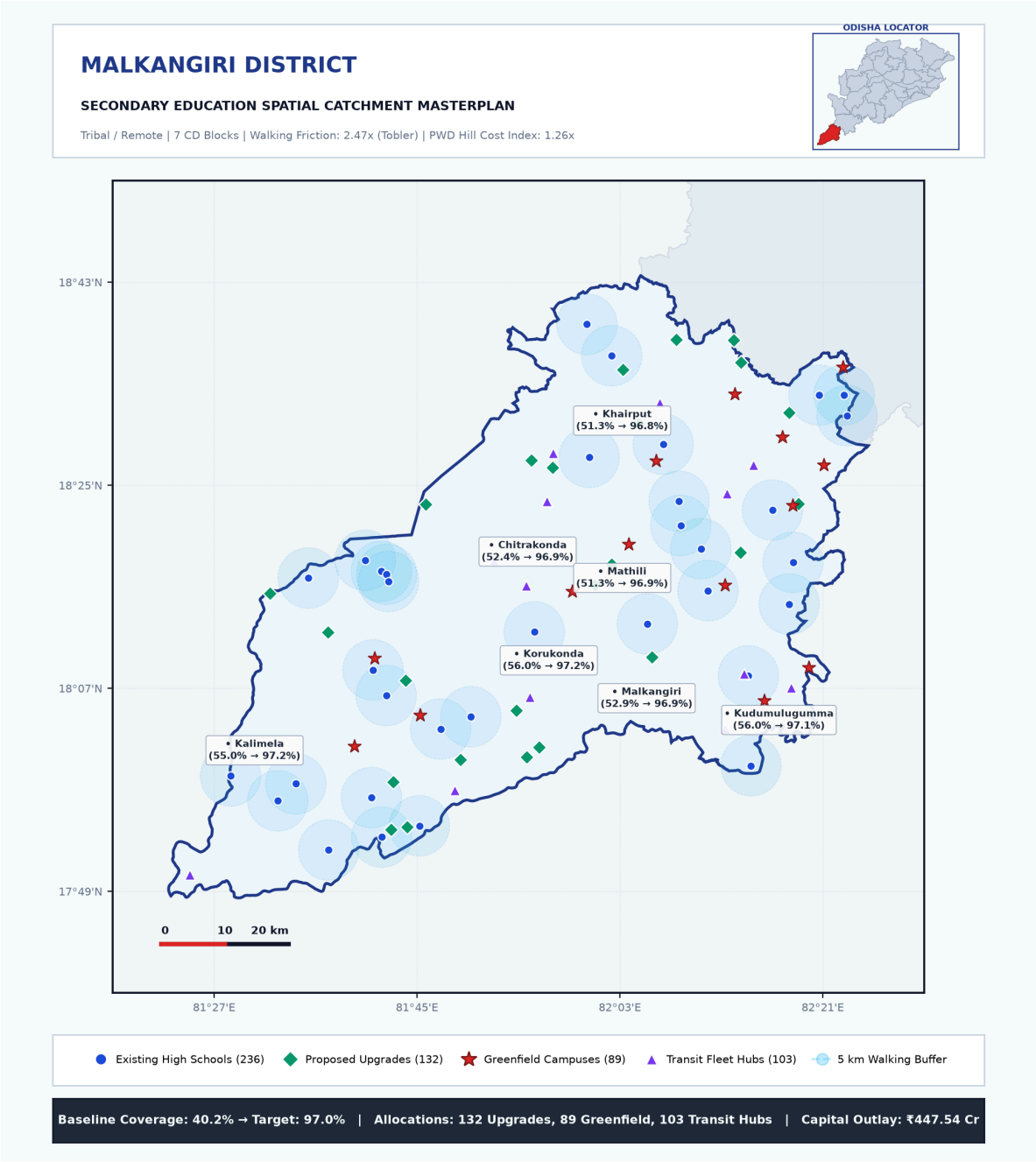
6.20 District Profile: Koraput (14 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	63.1%	55.1%	41.1%	35%
Proposed Upgrades	94	114	150	166
New Campuses	63	77	101	112
Transit Hubs	73	89	117	130
Est. Capital Outlay	₹88.8 Cr	₹197.8 Cr	₹508.1 Cr	₹1032.6 Cr

Operations Research Strategy: Koraput (Tribal / Hilly, Vulnerability: 93/100) comprises 14 CD blocks spanning 8,807 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.84, baseline secondary access is 41.1%. The masterplan allocates 150 high school upgrades, 101 greenfield campuses, 117 student transport hubs (104 mini-buses, 52 vans), 70 dedicated girls' hostels, and 704 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 508.13 Crores, raising universal secondary coverage to 97.0% across all 14 blocks.

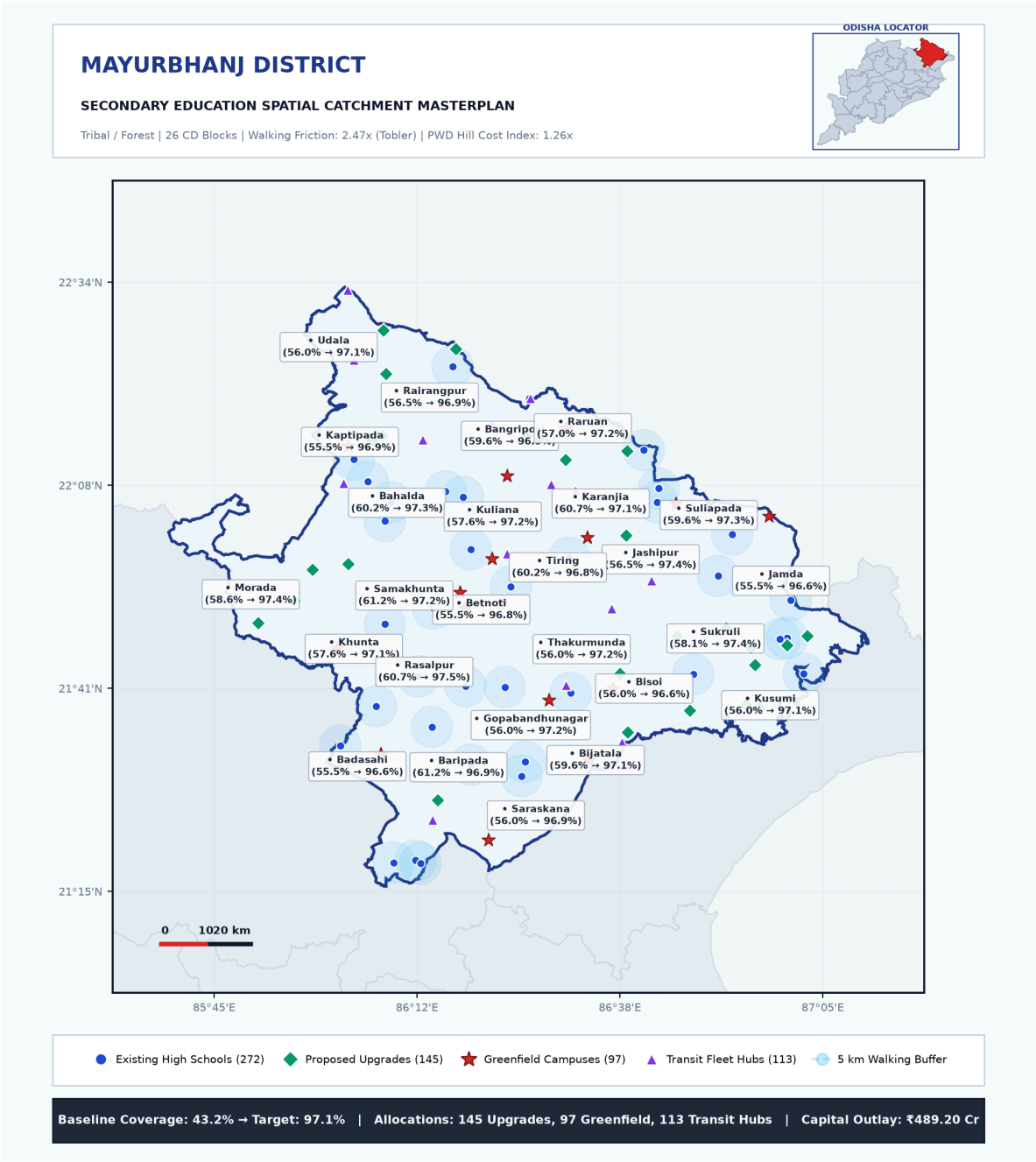
6.21 District Profile: Malkangiri (7 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	62.2%	54.2%	40.2%	35%
Proposed Upgrades	84	101	132	144
New Campuses	56	68	89	97
Transit Hubs	65	79	103	112
Est. Capital Outlay	₹79.1 Cr	₹174.9 Cr	₹447.5 Cr	₹894.4 Cr

Operations Research Strategy: Malkangiri (Tribal / Remote, Vulnerability: 96/100) comprises 7 CD blocks spanning 5,791 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.82, baseline secondary access is 40.2%. The masterplan allocates 132 high school upgrades, 89 greenfield campuses, 103 student transport hubs (84 mini-buses, 48 vans), 61 dedicated girls' hostels, and 620 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 447.54 Crores, raising universal secondary coverage to 97.0% across all 7 blocks.

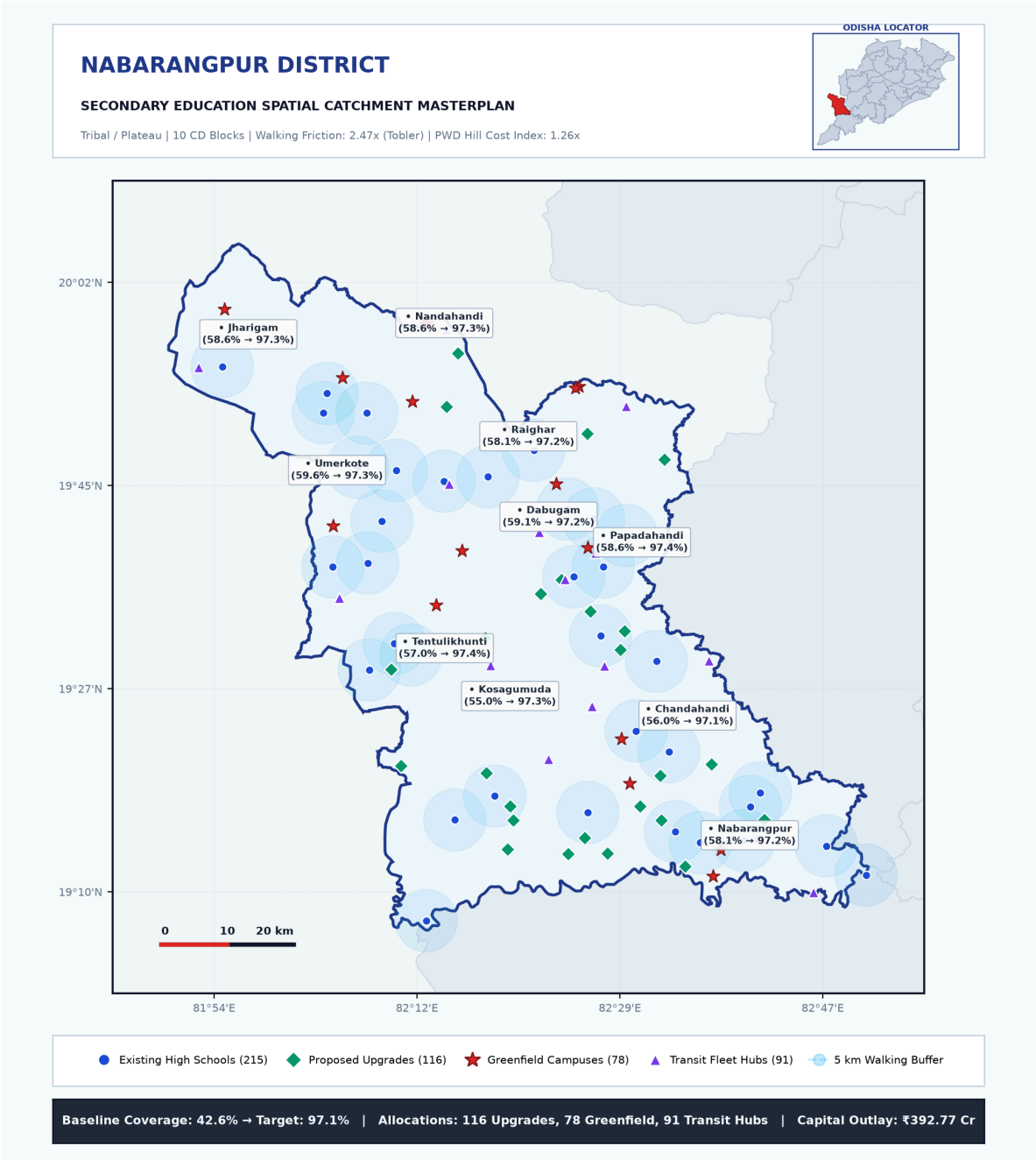
6.22 District Profile: Mayurbhanj (26 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	65.2%	57.2%	43.2%	35%
Proposed Upgrades	89	109	145	166
New Campuses	59	73	97	112
Transit Hubs	69	85	113	130
Est. Capital Outlay	₹83.6 Cr	₹188.2 Cr	₹489.2 Cr	₹1032.6 Cr

Operations Research Strategy: Mayurbhanj (Tribal / Forest, Vulnerability: 86/100) comprises 26 CD blocks spanning 10,418 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.88, baseline secondary access is 43.2%. The masterplan allocates 145 high school upgrades, 97 greenfield campuses, 113 student transport hubs (79 mini-buses, 27 vans), 67 dedicated girls' hostels, and 678 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 489.2 Crores, raising universal secondary coverage to 97.1% across all 26 blocks.

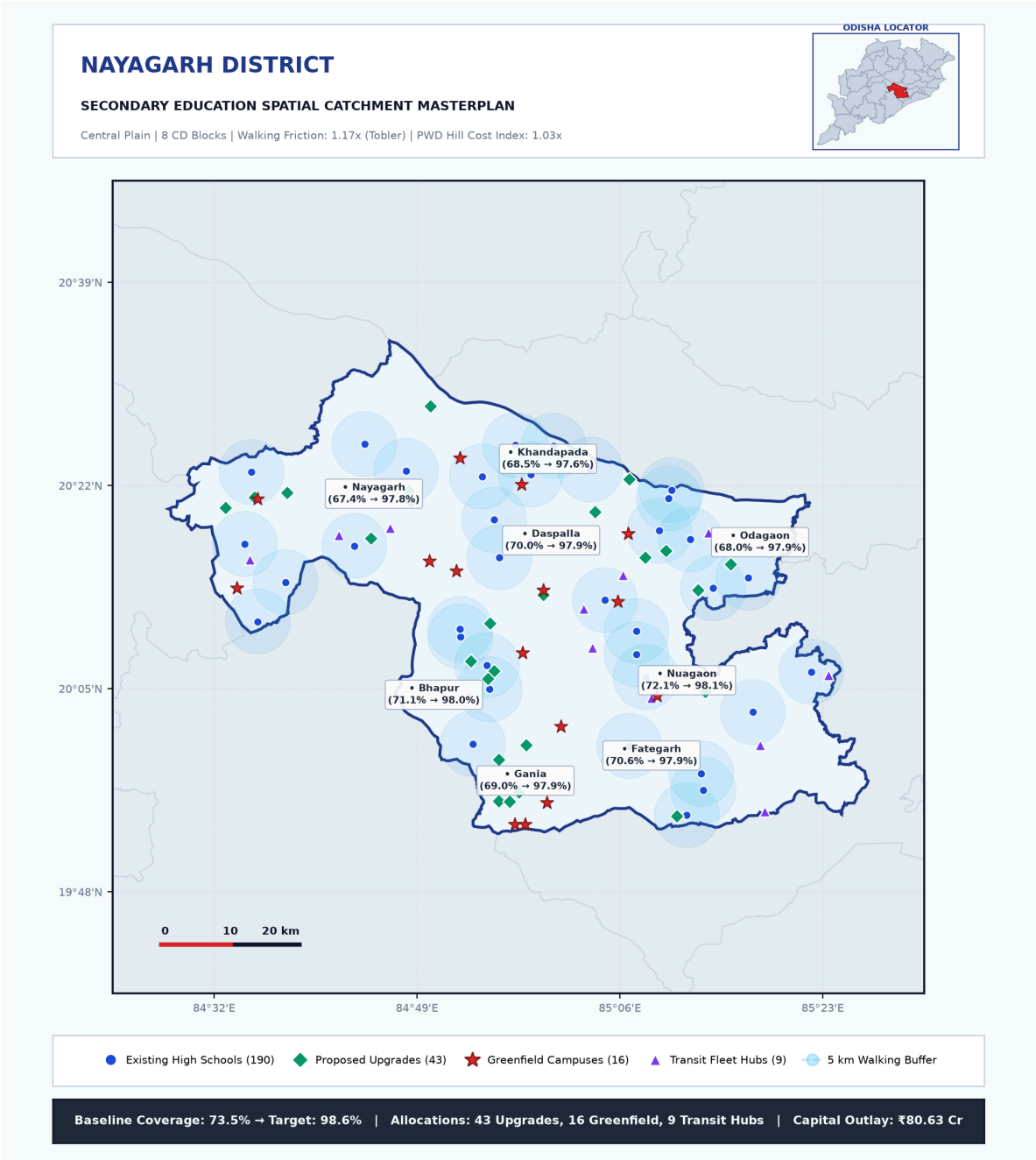
6.23 District Profile: Nabarangpur (10 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	64.6%	56.6%	42.6%	35%
Proposed Upgrades	71	88	116	132
New Campuses	48	59	78	88
Transit Hubs	56	68	91	103
Est. Capital Outlay	₹67.5 Cr	₹151.9 Cr	₹392.8 Cr	₹814.5 Cr

Operations Research Strategy: Nabarangpur (Tribal / Plateau, Vulnerability: 88/100) comprises 10 CD blocks spanning 5,291 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.85, baseline secondary access is 42.6%. The masterplan allocates 116 high school upgrades, 78 greenfield campuses, 91 student transport hubs (71 mini-buses, 36 vans), 54 dedicated girls' hostels, and 544 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 392.77 Crores, raising universal secondary coverage to 97.1% across all 10 blocks.

6.24 District Profile: Nayagarh (8 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	83.5%	77.5%	73.5%	60.5%
Proposed Upgrades	26	36	43	64
New Campuses	10	14	16	24
Transit Hubs	5	7	9	13
Est. Capital Outlay	₹13.9 Cr	₹35.3 Cr	₹80.6 Cr	₹217 Cr

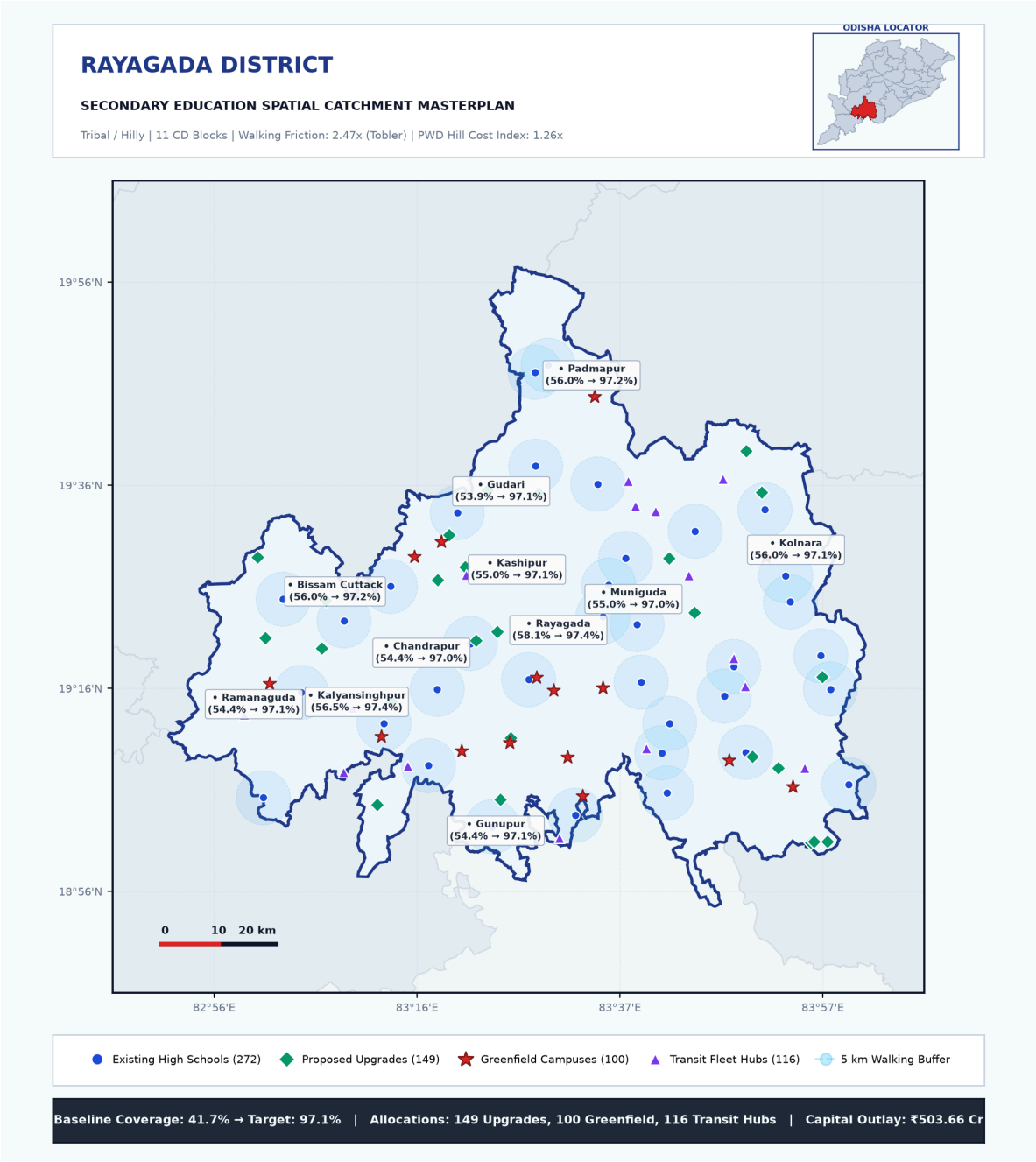
Operations Research Strategy: Nayagarh (Central Plain, Vulnerability: 65/100) comprises 8 CD blocks spanning 3,890 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 73.5%. The masterplan allocates 43 high school upgrades, 16 greenfield campuses, 9 student transport hubs (8 mini-buses, 8 vans), 0 dedicated girls' hostels, and 150 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 80.63 Crores, raising universal secondary coverage to 98.6% across all 8 blocks.

ODISHA LOCATOR



Operations Research Strategy: Nuapada (Drought-prone / Tribal, Vulnerability: 84/100) comprises 5 CD blocks spanning 3,852 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.87, baseline secondary access is 43.8%. The masterplan allocates 83 high school upgrades, 55 greenfield campuses, 64 student transport hubs (50 mini-buses, 28 vans), 38 dedicated girls' hostels, and 386 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 278.2 Crores, raising universal secondary coverage to 97.2% across all 5 blocks.

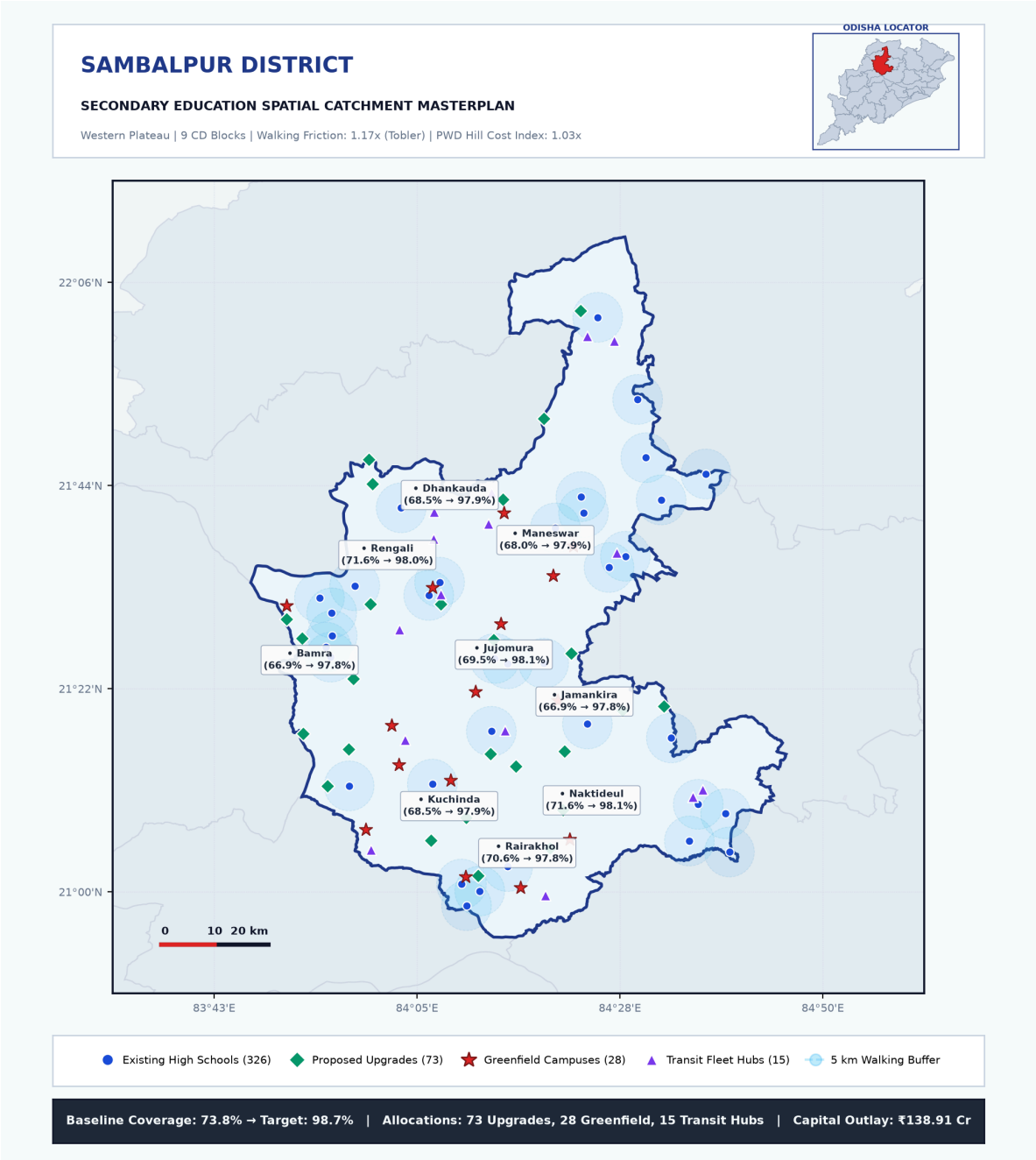
6.27 District Profile: Rayagada (11 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	63.7%	55.7%	41.7%	35%
Proposed Upgrades	92	113	149	166
New Campuses	62	76	100	112
Transit Hubs	72	88	116	130
Est. Capital Outlay	₹87.3 Cr	₹195.5 Cr	₹503.7 Cr	₹1032.6 Cr

Operations Research Strategy: Rayagada (Tribal / Hilly, Vulnerability: 91/100) comprises 11 CD blocks spanning 7,073 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.83, baseline secondary access is 41.7%. The masterplan allocates 149 high school upgrades, 100 greenfield campuses, 116 student transport hubs (103 mini-buses, 56 vans), 69 dedicated girls' hostels, and 698 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 503.66 Crores, raising universal secondary coverage to 97.1% across all 11 blocks.

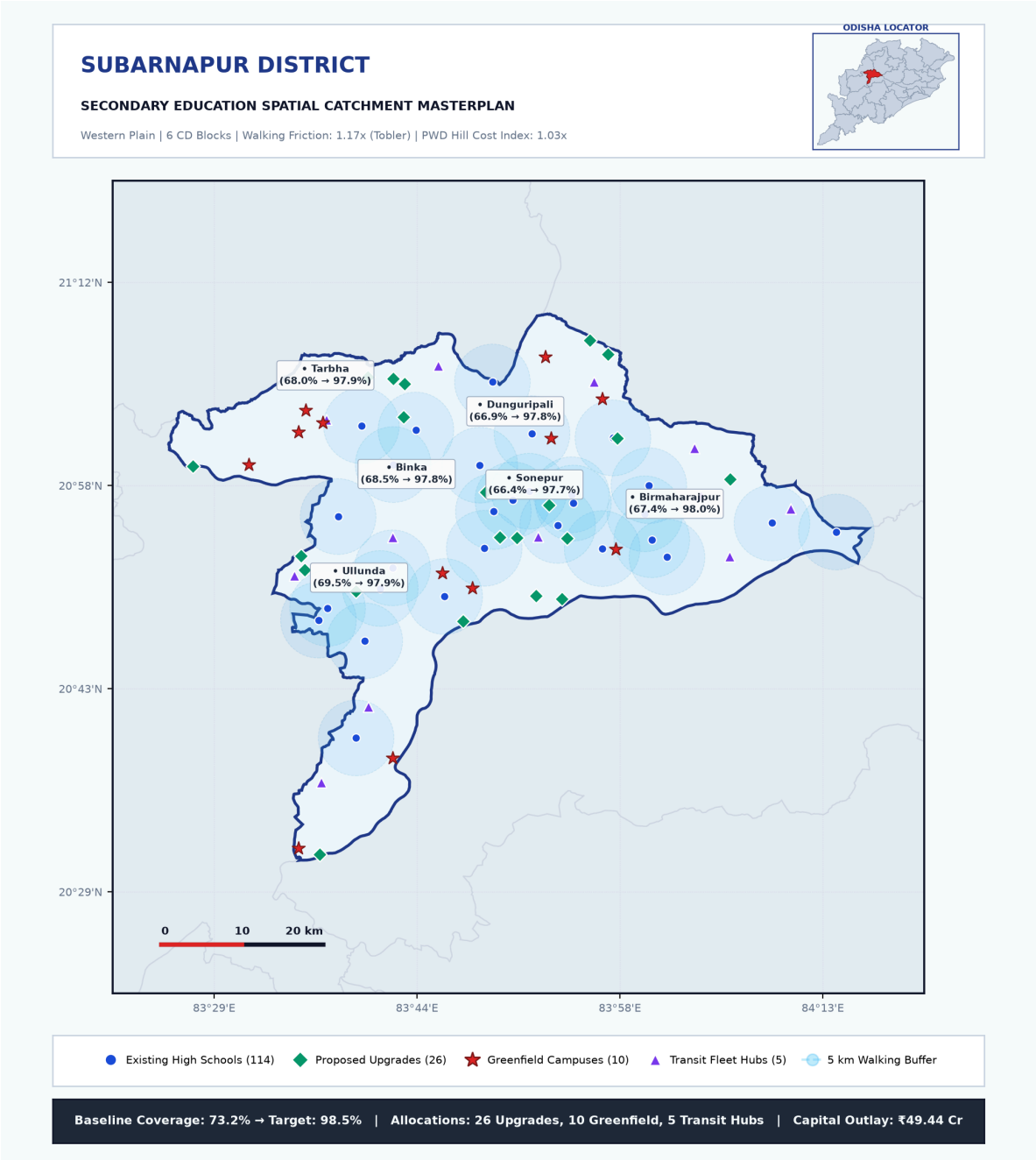
6.28 District Profile: Sambalpur (9 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	83.8%	77.8%	73.8%	60.8%
Proposed Upgrades	45	62	73	109
New Campuses	17	23	28	42
Transit Hubs	9	12	15	22
Est. Capital Outlay	₹23.9 Cr	₹59.4 Cr	₹138.9 Cr	₹375.1 Cr

Operations Research Strategy: Sambalpur (Western Plateau, Vulnerability: 64/100) comprises 9 CD blocks spanning 6,657 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.92, baseline secondary access is 73.8%. The masterplan allocates 73 high school upgrades, 28 greenfield campuses, 15 student transport hubs (10 mini-buses, 9 vans), 0 dedicated girls' hostels, and 258 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 138.91 Crores, raising universal secondary coverage to 98.7% across all 9 blocks.

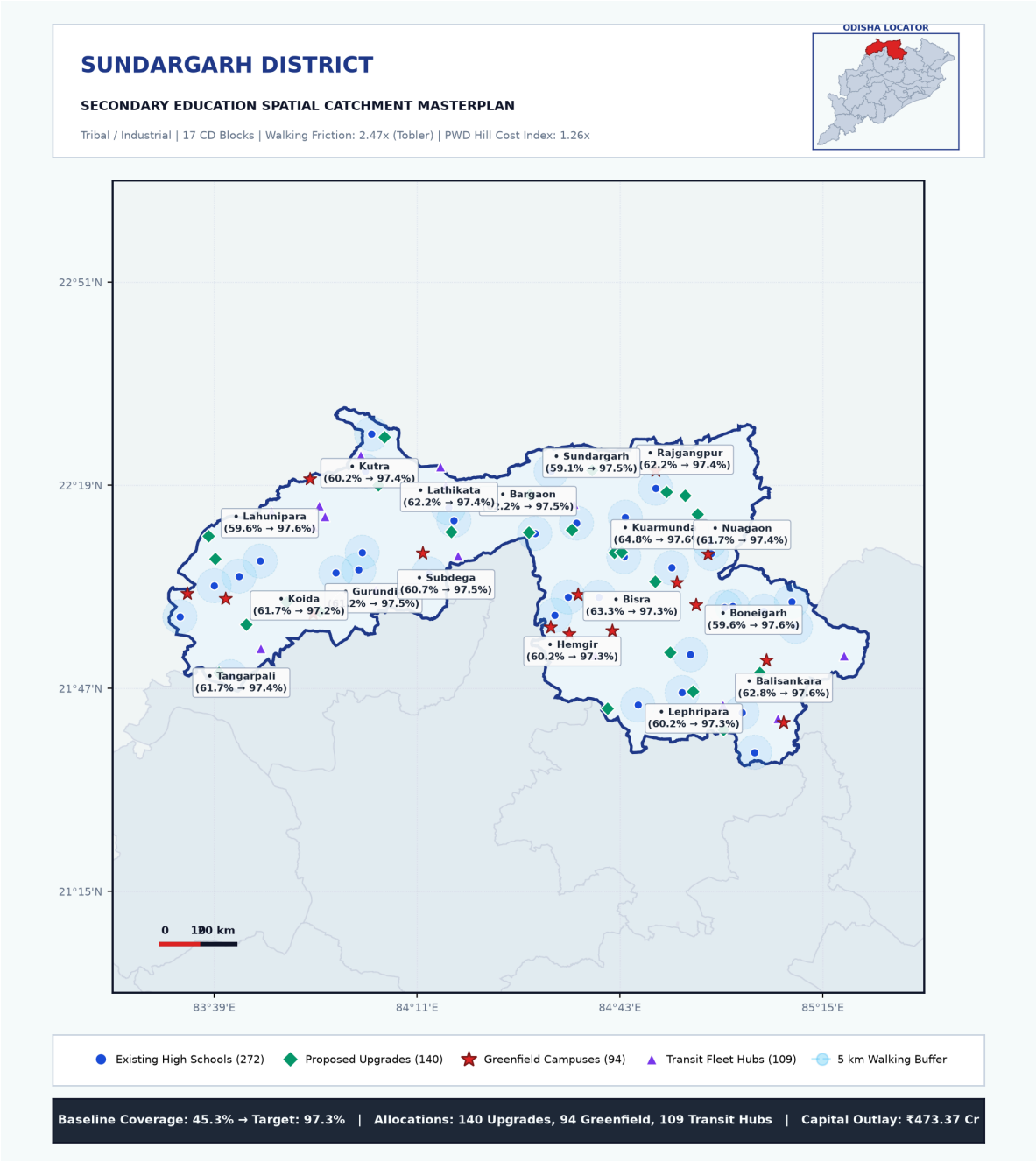
6.29 District Profile: Subarnapur (6 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	83.2%	77.2%	73.2%	60.2%
Proposed Upgrades	16	22	26	39
New Campuses	6	8	10	15
Transit Hubs	3	4	5	8
Est. Capital Outlay	₹8.4 Cr	₹20.8 Cr	₹49.4 Cr	₹134.1 Cr

Operations Research Strategy: Subarnapur (Western Plain, Vulnerability: 66/100) comprises 6 CD blocks spanning 2,337 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.92, baseline secondary access is 73.2%. The masterplan allocates 26 high school upgrades, 10 greenfield campuses, 5 student transport hubs (6 mini-buses, 6 vans), 0 dedicated girls' hostels, and 92 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 49.44 Crores, raising universal secondary coverage to 98.5% across all 6 blocks.

6.30 District Profile: Sundargarh (17 CD Blocks)



Metric	Primary (1 km)	Upper Prim. (3 km)	Secondary (5 km)	Higher Sec. (7 km)
Baseline Access	67.3%	59.3%	45.3%	35%
Proposed Upgrades	83	104	140	166
New Campuses	56	70	94	112
Transit Hubs	65	81	109	130
Est. Capital Outlay	₹78.8 Cr	₹180 Cr	₹473.4 Cr	₹1032.6 Cr

Operations Research Strategy: Sundargarh (Tribal / Industrial, Vulnerability: 79/100) comprises 17 CD blocks spanning 9,712 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.9, baseline secondary access is 45.3%. The masterplan allocates 140 high school upgrades, 94 greenfield campuses, 109 student transport hubs (76 mini-buses, 38 vans), 65 dedicated girls' hostels, and 656 subject teacher recruitments (25% tribal hardship allowance). Total estimated capital outlay is Rs. 473.37 Crores, raising universal secondary coverage to 97.3% across all 17 blocks.

7. Comprehensive 314-Block Analytical Appendix (Part 1 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Angul	Angul	143357	300	62	70.6%	98.2%	11	3	2	₹17.79 Cr
Angul	Athmallik	140813	298	61	71.1%	98.2%	11	3	2	₹17.79 Cr
Angul	Banarpal	140506	316	63	70%	98%	12	3	2	₹18.66 Cr
Angul	Chhendipada	167698	262	54	74.7%	98.4%	8	2	1	₹12.34 Cr
Angul	Kaniha	169497	269	65	69%	98%	10	3	2	₹16.91 Cr
Angul	Kishorenagar	165305	293	54	74.7%	98.4%	9	2	1	₹13.22 Cr
Angul	Pallahara	175959	300	55	74.2%	98.3%	10	2	1	₹14.09 Cr
Angul	Talcher	182339	246	61	71.1%	98.2%	9	2	1	₹13.22 Cr
Balangir	Agalpur	120402	182	77	62.8%	97.4%	8	2	1	₹12.34 Cr
Balangir	Balangir	106631	161	77	62.8%	97.2%	7	2	1	₹11.47 Cr
Balangir	Belpada	116998	185	72	65.4%	97.9%	8	2	1	₹12.34 Cr
Balangir	Deogaon	102734	159	81	60.7%	97.4%	8	2	1	₹12.34 Cr
Balangir	Gudvella	105389	142	79	61.7%	97.4%	7	2	1	₹11.47 Cr
Balangir	Khaprakhol	135237	154	78	62.2%	97.6%	7	2	1	₹11.47 Cr
Balangir	Loisingha	121541	177	73	64.8%	97.7%	8	2	1	₹12.34 Cr
Balangir	Muribahal	118582	150	81	60.7%	97%	7	2	1	₹11.47 Cr
Balangir	Patnagarh	130511	161	77	62.8%	97.2%	7	2	1	₹11.47 Cr
Balangir	Puintala	122061	154	83	59.6%	97.4%	8	2	1	₹12.34 Cr
Balangir	Saintala	133600	175	79	61.7%	97.7%	8	2	1	₹12.34 Cr
Balangir	Titilagarh	123453	149	74	64.3%	97.7%	6	2	1	₹10.59 Cr
Balangir	Turekela	102769	147	73	64.8%	97.4%	6	1	1	₹8.07 Cr
Balangir	Bangomunda	121099	161	72	65.4%	97.5%	7	2	1	₹11.47 Cr
Balasore	Bahanaga	175405	102	55	74.2%	98.2%	3	1	1	₹5.44 Cr
Balasore	Baleshwar	180529	108	50	76.8%	98.6%	3	1	1	₹5.44 Cr
Balasore	Baliapal	190077	125	47	78.4%	98%	3	1	1	₹5.44 Cr
Balasore	Basta	167305	112	55	74.2%	97.7%	3	1	1	₹5.44 Cr
Balasore	Bhograi	220584	122	56	73.7%	98.4%	4	1	1	₹6.32 Cr
Balasore	Jaleswar	174018	122	56	73.7%	98.4%	4	1	1	₹6.32 Cr
Balasore	Khaira	187636	101	50	76.8%	98.2%	2	1	1	₹4.57 Cr
Balasore	Nilagiri	221531	106	45	79.4%	98%	2	1	1	₹4.57 Cr
Balasore	Oupada	190766	120	48	77.8%	98.2%	3	1	1	₹5.44 Cr
Balasore	Remuna	222118	126	50	76.8%	98.4%	3	1	1	₹5.44 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 2 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Balasore	Simulia	173350	118	51	76.3%	97.8%	3	1	1	₹5.44 Cr
Balasore	Soro	197970	126	48	77.8%	97.9%	3	1	1	₹5.44 Cr
Bargarh	Ambabhona	116260	179	54	74.7%	98.3%	5	1	1	₹7.2 Cr
Bargarh	Attabira	124582	168	58	72.6%	98.3%	5	1	1	₹7.2 Cr
Bargarh	Bargarh	117171	182	57	73.2%	98%	6	1	1	₹8.07 Cr
Bargarh	Barpali	114231	158	60	71.6%	97.8%	5	1	1	₹7.2 Cr
Bargarh	Bhatli	109348	154	53	75.2%	98.4%	4	1	1	₹6.32 Cr
Bargarh	Bheden	111777	183	58	72.6%	98.3%	6	1	1	₹8.07 Cr
Bargarh	Bijepur	131753	169	54	74.7%	98.1%	5	1	1	₹7.2 Cr
Bargarh	Gaisilet	115401	171	63	70%	98%	6	1	1	₹8.07 Cr
Bargarh	Jharbandh	113345	154	53	75.2%	98.4%	4	1	1	₹6.32 Cr
Bargarh	Padmapur	124701	145	54	74.7%	98%	4	1	1	₹6.32 Cr
Bargarh	Paikmal	125412	194	55	74.2%	98.4%	6	1	1	₹8.07 Cr
Bargarh	Sohela	137081	150	60	71.6%	97.9%	5	1	1	₹7.2 Cr
Bhadrak	Basudevpur	186625	134	48	77.8%	98.1%	3	1	1	₹5.44 Cr
Bhadrak	Bhadrak	237990	125	49	77.3%	98.4%	3	1	1	₹5.44 Cr
Bhadrak	Bhandaripokhari	188106	142	43	80.4%	98.3%	3	1	1	₹5.44 Cr
Bhadrak	Bonth	226529	128	44	79.9%	98.3%	3	1	1	₹5.44 Cr
Bhadrak	Chandabali	240415	110	44	79.9%	98.7%	2	1	1	₹4.57 Cr
Bhadrak	Dhamnagar	222924	128	44	79.9%	98.3%	3	1	1	₹5.44 Cr
Bhadrak	Tihidi	216687	128	47	78.4%	98.2%	3	1	1	₹5.44 Cr
Boudh	Boudh	155598	342	80	61.2%	97.5%	17	5	3	₹28.38 Cr
Boudh	Harbhanga	160887	409	73	64.8%	97.7%	18	5	3	₹29.25 Cr
Boudh	Kantamal	138871	356	76	63.3%	97.6%	16	5	3	₹27.5 Cr
Cuttack	Athagarh	173112	84	38	83%	98.7%	1	1	1	₹3.69 Cr
Cuttack	Badamba	163438	108	41	81.5%	98%	2	1	1	₹4.57 Cr
Cuttack	Banki	170277	101	40	82%	98.8%	2	1	1	₹4.57 Cr
Cuttack	Banki-Dampada	173064	97	44	79.9%	98.3%	2	1	1	₹4.57 Cr
Cuttack	Baranga	201575	83	43	80.4%	98.5%	2	1	1	₹4.57 Cr
Cuttack	Cuttack Sadar	183252	86	45	79.4%	98%	2	1	1	₹4.57 Cr
Cuttack	Kantapada	188471	111	38	83%	98.2%	2	1	1	₹4.57 Cr
Cuttack	Mahanga	207051	90	36	84.1%	98.7%	1	1	1	₹3.69 Cr
Cuttack	Narasinghpur	196886	106	41	81.5%	98.3%	2	1	1	₹4.57 Cr
Cuttack	Niali	212771	100	47	78.4%	98.1%	2	1	1	₹4.57 Cr
Cuttack	Nischintakoili	199793	101	46	78.9%	98.4%	2	1	1	₹4.57 Cr
Cuttack	Salepur	190154	107	42	81%	98.6%	2	1	1	₹4.57 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 3 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Cuttack	Tangi-Choudwar	186037	105	46	78.9%	98.6%	2	1	1	₹4.57 Cr
Cuttack	Tigiria	195203	90	39	82.5%	98.2%	1	1	1	₹3.69 Cr
Deogarh	Barkote	101934	448	83	59.6%	97.4%	18	8	9	₹46.75 Cr
Deogarh	Reamal	99211	470	78	62.2%	97.6%	17	8	8	₹45.37 Cr
Deogarh	Tileibani	91845	515	84	59.1%	97.4%	21	9	10	₹53.36 Cr
Dhenkanal	Bhuban	154770	195	65	69%	97.9%	7	2	1	₹11.47 Cr
Dhenkanal	Dhenkanal	139516	169	59	72.1%	98.2%	6	1	1	₹8.07 Cr
Dhenkanal	Gandia	150897	183	59	72.1%	98.3%	6	1	1	₹8.07 Cr
Dhenkanal	Hindol	157776	178	57	73.2%	98%	6	1	1	₹8.07 Cr
Dhenkanal	Kamakhyanagar	144967	168	64	69.5%	98%	6	1	1	₹8.07 Cr
Dhenkanal	Kankadahad	135987	189	62	70.6%	98%	7	2	1	₹11.47 Cr
Dhenkanal	Odapada	152861	217	61	71.1%	98%	8	2	1	₹12.34 Cr
Dhenkanal	Parjang	160981	214	64	69.5%	98.1%	8	2	1	₹12.34 Cr
Gajapati	Gosani	80171	246	86	58.1%	97.5%	13	3	2	₹19.54 Cr
Gajapati	Gumma	86448	247	89	56.5%	97.2%	13	4	2	₹22.06 Cr
Gajapati	Kashinagar	74807	210	89	56.5%	97.2%	11	3	2	₹17.79 Cr
Gajapati	Mohana	85649	232	87	57.6%	97.3%	12	3	2	₹18.66 Cr
Gajapati	Nuagada	83762	196	92	55%	97.2%	11	3	2	₹17.79 Cr
Gajapati	R.Udayagiri	87453	211	94	53.9%	97.1%	12	3	2	₹18.66 Cr
Gajapati	Rayagada Block	85240	246	84	59.1%	97.3%	13	3	2	₹19.54 Cr
Ganjam	Aska	177702	143	63	70%	97.6%	5	1	1	₹7.2 Cr
Ganjam	Bellaguntha	170935	122	57	73.2%	97.9%	4	1	1	₹6.32 Cr
Ganjam	Bhanjanagar	138076	144	62	70.6%	98%	5	1	1	₹7.2 Cr
Ganjam	Buguda	180666	135	53	75.2%	98.2%	4	1	1	₹6.32 Cr
Ganjam	Chatrapur	157232	139	64	69.5%	97.9%	5	1	1	₹7.2 Cr
Ganjam	Chikiti	178339	135	54	74.7%	98.4%	4	1	1	₹6.32 Cr
Ganjam	Dharakote	179823	137	58	72.6%	98%	4	1	1	₹6.32 Cr
Ganjam	Digapahandi	150466	141	53	75.2%	97.9%	4	1	1	₹6.32 Cr
Ganjam	Ganjam	161909	116	57	73.2%	98.3%	4	1	1	₹6.32 Cr
Ganjam	Hinjilicut	144528	141	59	72.1%	98.1%	5	1	1	₹7.2 Cr
Ganjam	Jagannathprasad	166170	144	53	75.2%	98%	4	1	1	₹6.32 Cr
Ganjam	Kabisuryanagar	143235	146	55	74.2%	98%	4	1	1	₹6.32 Cr
Ganjam	Khallikote	148635	120	58	72.6%	97.7%	4	1	1	₹6.32 Cr
Ganjam	Kukudakhandi	179717	141	53	75.2%	97.9%	4	1	1	₹6.32 Cr
Ganjam	Patrapur	157605	116	61	71.1%	97.8%	4	1	1	₹6.32 Cr
Ganjam	Polasara	167096	133	57	73.2%	97.9%	4	1	1	₹6.32 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 4 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Ganjam	Purushottampur	177027	120	58	72.6%	97.7%	4	1	1	₹6.32 Cr
Ganjam	Rangeilunda	159601	125	55	74.2%	98.3%	4	1	1	₹6.32 Cr
Ganjam	Sanakhemundi	164010	132	55	74.2%	98.4%	4	1	1	₹6.32 Cr
Ganjam	Surada	183410	122	64	69.5%	98%	4	1	1	₹6.32 Cr
Ganjam	Sheragada	163582	121	60	71.6%	98%	4	1	1	₹6.32 Cr
Ganjam	Beguniapada	138708	139	61	71.1%	98.2%	5	1	1	₹7.2 Cr
Jagatsinghpur	Balikuda	133634	74	44	79.9%	97.7%	1	1	1	₹3.69 Cr
Jagatsinghpur	Biridi	133035	61	40	82%	97.4%	1	1	1	₹3.69 Cr
Jagatsinghpur	Erasama	148717	73	39	82.5%	98%	1	1	1	₹3.69 Cr
Jagatsinghpur	Jagatsinghpur	129914	70	39	82.5%	98.6%	1	1	1	₹3.69 Cr
Jagatsinghpur	Kujang	154742	80	39	82.5%	99%	1	1	1	₹3.69 Cr
Jagatsinghpur	Naugaon	142779	63	36	84.1%	99%	1	1	1	₹3.69 Cr
Jagatsinghpur	Raghunathpur	155697	68	41	81.5%	98.1%	1	1	1	₹3.69 Cr
Jagatsinghpur	Tirtol	130380	75	43	80.4%	97.9%	1	1	1	₹3.69 Cr
Jajpur	Badachana	188445	95	52	75.8%	97.6%	2	1	1	₹4.57 Cr
Jajpur	Bari	186823	97	50	76.8%	98.1%	2	1	1	₹4.57 Cr
Jajpur	Binjharpur	155325	107	47	78.4%	98.6%	2	1	1	₹4.57 Cr
Jajpur	Danagadi	194649	102	55	74.2%	98.2%	3	1	1	₹5.44 Cr
Jajpur	Dasarathpur	175187	106	52	75.8%	98%	3	1	1	₹5.44 Cr
Jajpur	Dharmasala	173323	105	44	79.9%	98.7%	2	1	1	₹4.57 Cr
Jajpur	Jajpur	194666	111	47	78.4%	97.9%	2	1	1	₹4.57 Cr
Jajpur	Korei	177608	95	47	78.4%	98.2%	2	1	1	₹4.57 Cr
Jajpur	Rasulpur	162206	94	48	77.8%	97.8%	2	1	1	₹4.57 Cr
Jajpur	Sukinda	192341	114	49	77.3%	97.9%	3	1	1	₹5.44 Cr
Jharsuguda	Jharsuguda	122896	163	50	76.8%	98.1%	4	1	1	₹6.32 Cr
Jharsuguda	Kirmira	131480	125	49	77.3%	98.4%	3	1	1	₹5.44 Cr
Jharsuguda	Kolabira	115503	156	47	78.4%	98.3%	4	1	1	₹6.32 Cr
Jharsuguda	Laikera	120974	155	55	74.2%	97.9%	5	1	1	₹7.2 Cr
Jharsuguda	Lakhanpur	101751	159	51	76.3%	98.2%	4	1	1	₹6.32 Cr
Kalahandi	Bhawanipatna	123146	191	81	60.7%	97.6%	9	2	1	₹13.22 Cr
Kalahandi	Dharamgarh	139209	187	85	58.6%	97.3%	10	2	1	₹14.09 Cr
Kalahandi	Golamunda	107376	210	89	56.5%	97.2%	11	3	2	₹17.79 Cr
Kalahandi	Jaipatna	129179	212	88	57%	97.3%	11	3	2	₹17.79 Cr
Kalahandi	Junagarh	133477	198	85	58.6%	97.1%	10	3	2	₹16.91 Cr
Kalahandi	Kalampur	123162	196	90	56%	97.2%	11	3	2	₹17.79 Cr
Kalahandi	Karlamunda	121596	239	83	59.6%	97.4%	12	3	2	₹18.66 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 5 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Kalahandi	Kesinga	113154	236	89	56.5%	97.1%	13	3	2	₹19.54 Cr
Kalahandi	Koksara	137098	207	88	57%	97.4%	11	3	2	₹17.79 Cr
Kalahandi	Lanjigarh	113394	233	85	58.6%	97.3%	12	3	2	₹18.66 Cr
Kalahandi	M.Rampur	139451	218	82	60.2%	97.3%	11	3	2	₹17.79 Cr
Kalahandi	Narla	122700	190	84	59.1%	97.2%	10	2	1	₹14.09 Cr
Kalahandi	Thuamul Rampur	122877	216	83	59.6%	97.5%	11	3	2	₹17.79 Cr
Kandhamal	Baliguda	64885	244	91	55.5%	97.1%	10	4	5	₹24.6 Cr
Kandhamal	Chakapad	64526	262	96	52.9%	97%	12	5	6	₹30.14 Cr
Kandhamal	Daringbadi	63384	288	99	51.3%	97%	14	6	7	₹35.67 Cr
Kandhamal	G.Udayagiri	59719	300	95	53.4%	97%	13	6	6	₹34.3 Cr
Kandhamal	K.Nuagaon	68577	277	94	53.9%	97%	12	5	6	₹30.14 Cr
Kandhamal	Khajuripada	54969	231	97	52.4%	96.8%	10	5	5	₹27.69 Cr
Kandhamal	Kotagarh	62360	285	91	55.5%	97.1%	12	5	6	₹30.14 Cr
Kandhamal	Phiringia	64550	236	91	55.5%	97.3%	10	4	5	₹24.6 Cr
Kandhamal	Phulbani	61100	301	96	52.9%	96.9%	14	6	7	₹35.67 Cr
Kandhamal	Raikia	52658	232	93	54.4%	96.9%	10	4	5	₹24.6 Cr
Kandhamal	Tikabali	56516	251	93	54.4%	97.1%	11	5	5	₹28.76 Cr
Kandhamal	Tumudibandha	67249	302	89	56.5%	97.3%	13	6	6	₹34.3 Cr
Kendrapara	Aul	166317	88	50	76.8%	98.2%	2	1	1	₹4.57 Cr
Kendrapara	Derabish	177289	114	50	76.8%	98.2%	3	1	1	₹5.44 Cr
Kendrapara	Garadpur	168136	93	48	77.8%	98%	2	1	1	₹4.57 Cr
Kendrapara	Kendrapara	180931	90	50	76.8%	97.7%	2	1	1	₹4.57 Cr
Kendrapara	Mahakalapada	173687	101	54	74.7%	98%	3	1	1	₹5.44 Cr
Kendrapara	Marsaghai	140673	92	57	73.2%	97.7%	3	1	1	₹5.44 Cr
Kendrapara	Pattamundai	158440	99	53	75.2%	98%	3	1	1	₹5.44 Cr
Kendrapara	Rajkanika	183267	107	56	73.7%	98.3%	3	1	1	₹5.44 Cr
Kendrapara	Rajnagar	152309	99	49	77.3%	98.2%	2	1	1	₹4.57 Cr
Kendujhar	Anandapur	154452	268	87	57.6%	97.2%	11	5	5	₹28.76 Cr
Kendujhar	Banspal	131616	263	76	63.3%	97.6%	9	4	4	₹23.22 Cr
Kendujhar	Champua	153731	268	87	57.6%	97.2%	11	5	5	₹28.76 Cr
Kendujhar	Ghasipura	150542	228	85	58.6%	97.4%	9	4	4	₹23.22 Cr
Kendujhar	Ghatgaon	153503	273	77	62.8%	97.6%	10	4	5	₹24.6 Cr
Kendujhar	Harichandanpur	136943	269	78	62.2%	97.5%	10	4	5	₹24.6 Cr
Kendujhar	Hatadihi	127267	267	79	61.7%	97.6%	10	4	5	₹24.6 Cr
Kendujhar	Jhumpura	131453	223	76	63.3%	97.4%	8	3	4	₹19.06 Cr
Kendujhar	Joda	129410	280	86	58.1%	97.4%	11	5	5	₹28.76 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 6 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Kendujhar	Keonjhar	158599	238	83	59.6%	97.5%	9	4	4	₹23.22 Cr
Kendujhar	Patna	122600	278	82	60.2%	97.4%	11	5	5	₹28.76 Cr
Kendujhar	Saharapada	157825	222	87	57.6%	97.4%	9	4	4	₹23.22 Cr
Kendujhar	Telkoi	135873	217	79	61.7%	97.7%	8	3	4	₹19.06 Cr
Khordha	Balianta	215153	83	36	84.1%	98.8%	1	1	1	₹3.69 Cr
Khordha	Balipatna	255796	108	41	81.5%	98%	2	1	1	₹4.57 Cr
Khordha	Banapur	228283	105	37	83.6%	98.8%	2	1	1	₹4.57 Cr
Khordha	Begunia	206213	89	39	82.5%	98.3%	1	1	1	₹3.69 Cr
Khordha	Bhubaneswar	214097	84	37	83.6%	98.1%	1	1	1	₹3.69 Cr
Khordha	Bolagarh	202540	95	40	82%	98.8%	2	1	1	₹4.57 Cr
Khordha	Chilika	233425	87	37	83.6%	98.7%	1	1	1	₹3.69 Cr
Khordha	Jatni	229516	94	32	86.2%	98.2%	1	1	1	₹3.69 Cr
Khordha	Khordha	200558	102	32	86.2%	99.1%	1	1	1	₹3.69 Cr
Khordha	Tangi	217000	84	37	83.6%	98.1%	1	1	1	₹3.69 Cr
Koraput	Bandhugaon	88360	217	95	53.4%	97.2%	10	4	5	₹24.6 Cr
Koraput	Boipariguda	86506	216	98	51.8%	97.1%	10	4	5	₹24.6 Cr
Koraput	Borigumma	101612	251	88	57%	97.1%	10	4	5	₹24.6 Cr
Koraput	Damanjodi	99617	243	96	52.9%	97%	11	5	5	₹28.76 Cr
Koraput	Dasamantapur	100084	219	98	51.8%	96.9%	10	4	5	₹24.6 Cr
Koraput	Jeypore	109239	236	96	52.9%	97.1%	11	5	5	₹28.76 Cr
Koraput	Kotpad	85118	241	93	54.4%	96.9%	10	5	5	₹27.69 Cr
Koraput	Kundra	86123	213	89	56.5%	97.1%	9	4	4	₹23.22 Cr
Koraput	Lamtaput	91982	200	89	56.5%	97.4%	8	4	4	₹22.15 Cr
Koraput	Laxmipur	94708	218	94	53.9%	97%	10	4	5	₹24.6 Cr
Koraput	Nandapur	111455	212	89	56.5%	97.3%	9	4	4	₹23.22 Cr
Koraput	Narayanpatna	88823	235	94	53.9%	97.1%	10	4	5	₹24.6 Cr
Koraput	Pattangi	94982	204	95	53.4%	97.2%	9	4	4	₹23.22 Cr
Koraput	Semiliguda	100229	237	98	51.8%	97%	11	5	5	₹28.76 Cr
Malkangiri	Chitrakonda	81685	340	97	52.4%	96.9%	16	7	8	₹41.21 Cr
Malkangiri	Kalimela	92241	425	92	55%	97.2%	19	8	9	₹47.82 Cr
Malkangiri	Khairput	85546	347	99	51.3%	96.8%	16	7	8	₹41.21 Cr
Malkangiri	Korukonda	87738	363	90	56%	97.2%	15	7	7	₹39.83 Cr
Malkangiri	Kudumulugumma	91536	361	90	56%	97.1%	15	7	7	₹39.83 Cr
Malkangiri	Malkangiri	89943	363	96	52.9%	96.9%	17	7	8	₹42.29 Cr
Malkangiri	Mathili	95634	338	99	51.3%	96.9%	16	7	8	₹41.21 Cr
Mayurbhanj	Badasahi	103198	112	91	55.5%	96.6%	4	2	2	₹11.07 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 7 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Mayurbhanj	Bahalda	101904	119	82	60.2%	97.3%	4	2	2	₹11.07 Cr
Mayurbhanj	Bangriposi	104235	116	83	59.6%	96.9%	4	2	2	₹11.07 Cr
Mayurbhanj	Baripada	111405	121	80	61.2%	96.9%	4	2	2	₹11.07 Cr
Mayurbhanj	Betnoti	88573	107	91	55.5%	96.8%	4	2	2	₹11.07 Cr
Mayurbhanj	Bijatala	107986	138	83	59.6%	97.1%	5	2	2	₹12.15 Cr
Mayurbhanj	Bisoi	86962	118	90	56%	96.6%	5	2	2	₹12.15 Cr
Mayurbhanj	Gopabandhunagar	100159	130	90	56%	97.2%	5	2	2	₹12.15 Cr
Mayurbhanj	Jamda	82603	128	91	55.5%	96.6%	5	2	2	₹12.15 Cr
Mayurbhanj	Jashipur	101663	115	89	56.5%	97.4%	5	2	2	₹12.15 Cr
Mayurbhanj	Kaptipada	85732	109	91	55.5%	96.9%	4	2	2	₹11.07 Cr
Mayurbhanj	Karanjia	90294	124	81	60.7%	97.1%	4	2	2	₹11.07 Cr
Mayurbhanj	Khunta	88295	131	87	57.6%	97.1%	5	2	2	₹12.15 Cr
Mayurbhanj	Kuliana	96579	114	87	57.6%	97.2%	4	2	2	₹11.07 Cr
Mayurbhanj	Kusumi	85059	135	90	56%	97.1%	5	2	2	₹12.15 Cr
Mayurbhanj	Morada	82479	114	85	58.6%	97.4%	4	2	2	₹11.07 Cr
Mayurbhanj	Rairangpur	89992	128	89	56.5%	96.9%	5	2	2	₹12.15 Cr
Mayurbhanj	Raruan	94810	124	88	57%	97.2%	5	2	2	₹12.15 Cr
Mayurbhanj	Samakhunta	108051	107	80	61.2%	97.2%	4	1	2	₹7.99 Cr
Mayurbhanj	Saraskana	106640	124	90	56%	96.9%	5	2	2	₹12.15 Cr
Mayurbhanj	Sukruli	86081	110	86	58.1%	97.4%	4	2	2	₹11.07 Cr
Mayurbhanj	Suliapada	105522	137	83	59.6%	97.3%	5	2	2	₹12.15 Cr
Mayurbhanj	Thakurmunda	88483	137	90	56%	97.2%	6	2	3	₹13.52 Cr
Mayurbhanj	Tiring	105057	108	82	60.2%	96.8%	4	1	2	₹7.99 Cr
Mayurbhanj	Udala	100421	119	90	56%	97.1%	5	2	2	₹12.15 Cr
Mayurbhanj	Rasalpur	107513	138	81	60.7%	97.5%	5	2	2	₹12.15 Cr
Nabarangpur	Chandahandi	132203	279	90	56%	97.1%	12	5	6	₹30.14 Cr
Nabarangpur	Dabugam	115334	217	84	59.1%	97.2%	8	4	4	₹22.15 Cr
Nabarangpur	Jharigam	138316	231	85	58.6%	97.3%	9	4	4	₹23.22 Cr
Nabarangpur	Kosagumuda	127787	238	92	55%	97.3%	10	4	5	₹24.6 Cr
Nabarangpur	Nabarangpur	120586	284	86	58.1%	97.2%	11	5	5	₹28.76 Cr
Nabarangpur	Nandahandi	124341	233	85	58.6%	97.3%	9	4	4	₹23.22 Cr
Nabarangpur	Papadahandi	136665	259	85	58.6%	97.4%	10	4	5	₹24.6 Cr
Nabarangpur	Raighar	140317	231	86	58.1%	97.2%	9	4	4	₹23.22 Cr
Nabarangpur	Tentulikhunti	105505	221	88	57%	97.4%	9	4	4	₹23.22 Cr
Nabarangpur	Umerkote	132792	262	83	59.6%	97.3%	10	4	5	₹24.6 Cr
Nayagarh	Bhapur	129779	161	61	71.1%	98%	5	1	1	₹7.2 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 8 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Nayagarh	Daspalla	132331	182	63	70%	97.9%	7	2	1	₹11.47 Cr
Nayagarh	Fategarh	102995	148	62	70.6%	97.9%	5	1	1	₹7.2 Cr
Nayagarh	Gania	114930	195	65	69%	97.9%	7	2	1	₹11.47 Cr
Nayagarh	Khandapada	125827	184	66	68.5%	97.6%	7	2	1	₹11.47 Cr
Nayagarh	Nayagarh	109493	192	68	67.4%	97.8%	8	2	1	₹12.34 Cr
Nayagarh	Nuagaon	106852	152	59	72.1%	98.1%	5	1	1	₹7.2 Cr
Nayagarh	Odagaon	110165	173	67	68%	97.9%	7	2	1	₹11.47 Cr
Nuapada	Boden	117740	320	83	59.6%	97.5%	12	5	6	₹30.14 Cr
Nuapada	Khariar	139328	372	89	56.5%	97.2%	16	7	8	₹41.21 Cr
Nuapada	Komna	130159	314	88	57%	97.4%	13	6	6	₹34.3 Cr
Nuapada	Nuapada	113536	373	86	58.1%	97.4%	15	7	7	₹39.83 Cr
Nuapada	Sinapali	119686	325	85	58.6%	97.4%	13	6	6	₹34.3 Cr
Puri	Astaranga	173154	109	46	78.9%	97.9%	2	1	1	₹4.57 Cr
Puri	Brahmagiri	159390	119	42	81%	98.4%	2	1	1	₹4.57 Cr
Puri	Delanga	160030	114	43	80.4%	98.5%	2	1	1	₹4.57 Cr
Puri	Gop	161804	117	47	78.4%	98.5%	3	1	1	₹5.44 Cr
Puri	Kanas	173119	114	49	77.3%	97.9%	3	1	1	₹5.44 Cr
Puri	Kakatpur	151688	103	46	78.9%	98.1%	2	1	1	₹4.57 Cr
Puri	Krushnaprasad	135441	117	45	79.4%	98.7%	3	1	1	₹5.44 Cr
Puri	Nimapada	139402	118	42	81%	98.5%	2	1	1	₹4.57 Cr
Puri	Pipili	176273	111	40	82%	98.1%	2	1	1	₹4.57 Cr
Puri	Puri Sadar	169740	123	45	79.4%	98.5%	3	1	1	₹5.44 Cr
Puri	Satyabadi	153589	120	42	81%	98.2%	2	1	1	₹4.57 Cr
Rayagada	Bissam Cuttack	79550	274	90	56%	97.2%	12	5	6	₹30.14 Cr
Rayagada	Chandrapur	80732	296	93	54.4%	97%	13	6	6	₹34.3 Cr
Rayagada	Gudari	94187	250	94	53.9%	97.1%	11	5	5	₹28.76 Cr
Rayagada	Gunupur	84747	317	93	54.4%	97.1%	14	6	7	₹35.67 Cr
Rayagada	Kashipur	100402	317	92	55%	97.1%	14	6	7	₹35.67 Cr
Rayagada	Kolnara	87806	249	90	56%	97.1%	10	5	5	₹27.69 Cr
Rayagada	Muniguda	97596	322	92	55%	97%	14	6	7	₹35.67 Cr
Rayagada	Padmapur	86612	292	90	56%	97.2%	12	5	6	₹30.14 Cr
Rayagada	Ramanaguda	91820	282	93	54.4%	97.1%	12	5	6	₹30.14 Cr
Rayagada	Rayagada	100097	287	86	58.1%	97.4%	12	5	6	₹30.14 Cr
Rayagada	Kalyansinghpur	91690	306	89	56.5%	97.4%	13	6	6	₹34.3 Cr
Sambalpur	Bamra	127084	268	69	66.9%	97.8%	11	3	2	₹17.79 Cr
Sambalpur	Dhankauda	123790	253	66	68.5%	97.9%	10	3	1	₹16.61 Cr

7. Comprehensive 314-Block Analytical Appendix (Part 9 of 9)

Complete micro-demographic and secondary infrastructure allocation matrix across all 314 Community Development (CD) Blocks in Odisha.

District	CD Block	Population	Habs	Vuln.	Base Cov.	Target Cov.	Upgrades	New	Transit	Est. Outlay
Sambalpur	Jamankira	126372	240	69	66.9%	97.8%	10	3	1	₹16.61 Cr
Sambalpur	Jujomura	113441	256	64	69.5%	98.1%	10	3	1	₹16.61 Cr
Sambalpur	Kuchinda	127882	240	66	68.5%	97.9%	9	2	1	₹13.22 Cr
Sambalpur	Maneswar	128920	226	67	68%	97.9%	9	2	1	₹13.22 Cr
Sambalpur	Naktideul	119505	255	60	71.6%	98.1%	9	2	1	₹13.22 Cr
Sambalpur	Rairakhol	127856	221	62	70.6%	97.8%	8	2	1	₹12.34 Cr
Sambalpur	Rengali	126012	235	60	71.6%	98%	8	2	1	₹12.34 Cr
Subarnapur	Binka	116938	154	66	68.5%	97.8%	6	1	1	₹8.07 Cr
Subarnapur	Birmaharajpur	87669	126	68	67.4%	98%	5	1	1	₹7.2 Cr
Subarnapur	Dunguripali	106319	134	69	66.9%	97.8%	5	1	1	₹7.2 Cr
Subarnapur	Sonepur	104300	123	70	66.4%	97.7%	5	1	1	₹7.2 Cr
Subarnapur	Tarbha	89226	135	67	68%	97.9%	5	1	1	₹7.2 Cr
Subarnapur	Ullunda	106887	129	64	69.5%	97.9%	5	1	1	₹7.2 Cr
Sundargarh	Bargaon	137028	213	78	62.2%	97.5%	8	3	4	₹19.06 Cr
Sundargarh	Bisra	111271	188	76	63.3%	97.3%	6	3	3	₹16.61 Cr
Sundargarh	Boneigarh	115696	208	83	59.6%	97.6%	8	3	4	₹19.06 Cr
Sundargarh	Gurundia	110317	194	80	61.2%	97.5%	7	3	3	₹17.69 Cr
Sundargarh	Hemgir	133436	190	82	60.2%	97.3%	7	3	3	₹17.69 Cr
Sundargarh	Koida	116646	159	79	61.7%	97.2%	6	2	3	₹13.52 Cr
Sundargarh	Kuarmunda	137134	212	73	64.8%	97.6%	7	3	3	₹17.69 Cr
Sundargarh	Kutra	106811	177	82	60.2%	97.4%	7	3	3	₹17.69 Cr
Sundargarh	Lahunipara	107835	213	83	59.6%	97.6%	8	3	4	₹19.06 Cr
Sundargarh	Lathikata	132770	163	78	62.2%	97.4%	6	2	3	₹13.52 Cr
Sundargarh	Lephrupara	122229	167	82	60.2%	97.3%	6	3	3	₹16.61 Cr
Sundargarh	Nuagaon	136901	174	79	61.7%	97.4%	6	3	3	₹16.61 Cr
Sundargarh	Rajgangpur	124594	171	78	62.2%	97.4%	6	2	3	₹13.52 Cr
Sundargarh	Subdega	116187	171	81	60.7%	97.5%	6	3	3	₹16.61 Cr
Sundargarh	Sundargarh	120856	196	84	59.1%	97.5%	8	3	4	₹19.06 Cr
Sundargarh	Tangarpali	112972	166	79	61.7%	97.4%	6	2	3	₹13.52 Cr
Sundargarh	Balisankara	113172	192	77	62.8%	97.6%	7	3	3	₹17.69 Cr

8. Implementation Roadmap, Staff Retention & Governance

To achieve universal secondary schooling equity efficiently across all 314 blocks, the masterplan establishes a structured 3-phase rollout:

- Phase 1: High Vulnerability & Remote Tribal Corridors (Years 1–2 | ₹2,703.8 Cr / 45%)**
 - Priority Districts:** Malkangiri, Koraput, Rayagada, Kandhamal, Gajapati, Nabarangpur, Mayurbhanj, Nuapada, Kalahandi.
 - Interventions:** Fast-track construction of 588 dedicated girls' hostels, deployment of 1,172 student transit hubs (1032 mini-buses, 608 feeder vans), and 4,115 subject teacher recruitments.
- Phase 2: Mineral Belts, Plateaus & Western Agrarian Plains (Years 3–4 | ₹2,102.9 Cr / 35%)**
 - Priority Districts:** Kendujhar, Sundargarh, Deogarh, Balangir, Boudh, Sambalpur, Bargarh, Sonepur, Angul, Dhenkanal, Nayagarh.
 - Interventions:** Construction of greenfield high schools in dense mining periphery settlements and 3,200 subject teacher recruitments.
- Phase 3: Coastal Deltas, Disaster Retrofits & Urban Consolidation (Year 5 | ₹1,201.7 Cr / 20%)**
 - Priority Districts:** Khordha, Cuttack, Puri, Jagatsinghpur, Kendrapara, Jajpur, Bhadrak, Baleshwar, Ganjam, Jharsuguda.
 - Interventions:** 396 cyclone-resilient structural retrofits, advanced STEM smart-lab installations, and digital classroom integration.

Special Tribal Teacher Retention Cadre & PESA Land Protocols

- Tribal Hardship Allowance & Retention Bond:** To resolve the historic 9,144 specialized teacher recruitment bottleneck in Scheduled areas, teachers posted in Tobler friction zones (> 1.8x) will receive a **25% Remote Area Allowance** paired with a mandatory 3-year rural posting bond.
- PESA & FRA Fast-Track Land Resolution:** Greenfield school site selections prioritize unencumbered revenue wasteland with Gram Sabha consent under Section 4(i) of PESA to prevent construction delays.
- Calibrated Transit Opex:** Annual operating expenditure is calibrated at **₹68 Cr/year** (Mini-Bus at ₹4.80L/yr, Van at ₹3.00L/yr) including Mission Shakti female chaperone honorariums.
- Mathematical Edge Governance:** Disbursal of state capital grants is linked to GPS verification confirming optimal submodular catchment overlap.